FIELDCOMMAND

Incident Management System v1.0

INSTALLATION GUIDE

ASUS RT-BE58 Go · UniFi Switch Flex 2.5G · Raspberry Pi 5 · Pironman MAX 5

ICS/NIMS All-Hazards · Amateur Radio EMCOMM · July 23, 2026

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1. Overview & What's Included

FieldCommand is a complete off-grid emergency communications server built on two Raspberry Pi 5 units — one running all 32 EmComm tools, one dedicated to the AMPRNet / 44Net gateway. A cellular modem is the primary WAN source with a satellite dish as automatic failover, both managed by the ASUS RT-BE58 Go Wi-Fi 7 router. Two additional ASUS RT-BE58 Go routers extend EMCOMM-NET as AiMesh nodes for large venues. Any phone, tablet, or laptop connects to EMCOMM-NET and reaches all tools at http://192.168.50.1 — no internet, no app, no per-device setup required. When WAN is available, NWS alerts, APRS-IS, and HF propagation data activate automatically.

SAR Network Operations: The Starcom Net Logger includes a dedicated SAR Net mode (Search & Rescue) accessible directly from the Starcom / Public Safety dashboard. It provides a SAR-specific check-in log with unit type tracking, zone assignments, and resource status. The Starcom Resource Map provides the tactical overlay for SAR unit positions. These features are integrated into the main FieldCommand platform — no separate module is required.

COMPONENT	DESCRIPTION	COMPONENT	DESCRIPTION
30 Web Pages	Full dashboard, net logs, ICS forms, maps, roster, cheat sheets	Dead Man's Switch	Per-net inactivity monitor with warning and trigger states
Net Control Logger	Amateur + Starcom nets, FCC autofill, ICS-309 export	Pre-Flight Checklist	GO/CAUTION/NO-GO deployment readiness check
ICS Platform	Command, Operations, Planning, Logistics, Finance sections	HF Propagation	Solar indices, band conditions, A/K-index
APRS Tactical Map	Graywolf + YAAC merged, offline tiles, overlays	Repeater Database	RepeaterBook CSV, filter by band, mode, ARES affiliation
Member Roster	your organization directory with certs, equipment, activations	Reference Library	Upload and serve field reference docs across EMCOMM-NET
NTS Radiogram	ARRL-formatted radiogram generator with traffic log	Kiwix Library	WikiMed, Wikipedia, iFixit — offline at port 8081
ICS-213/214/309	Fillable ICS forms with print output and incident archiving	FCC Callsign Lookup	~800K licensees — instant offline search
Winlink Form Import	Import Winlink Express XML forms to incident archive	Pat Winlink	Browser-based backup Winlink client at port 8090
JS8Call Integration	Dashboard card opens JS8Call web API on Windows laptop	ICS Planning P	Interactive 15-phase planning cycle guide

2. Hardware Requirements

Supported Hardware — Raspberry Pi Servers

FieldCommand v1.0 uses two Raspberry Pi 5 units: one as the FieldCommand application server and one as the dedicated AMPRNet / 44Net gateway. Keeping them separate means the gateway can be rebooted, reconfigured, or upgraded without affecting the FieldCommand server, net logs, or active incidents.

UNIT	HARDWARE	STORAGE	ROLE
FieldCommand Server	Raspberry Pi 5 — 16 GB RAM / Pironman MAX 5 tower enclosure (dual M.2 NVMe, OLED, active cooling)	2× 1 TB NVMe SSD / configured as RAID 1	Runs all FieldCommand services: web server, FCC database, net logging, ICS platform, APRS, Kiwix, Winlink, maps, and health monitor. Static IP: 192.168.50.1
44Net Gateway	Raspberry Pi 5 — 16 GB RAM / Argon NEO 5 M.2 BRED case (compact, passive-cooled, M.2 SATA slot) / Pi OS Desktop (64-bit)	256 GB M.2 SATA SSD (OS + WireGuard + routing + status service)	Dedicated AMPRNet / 44Net gateway. Runs WireGuard tunnel to amprgw.ampr.org, ip routing between EMCOMM-NET and 44.0.0.0/8, and a status page at 192.168.50.2. Completely independent from the FieldCommand server.

Network Hardware

Wi-Fi is provided by the ASUS RT-BE58 Go travel router — neither Pi runs a Wi-Fi hotspot. The UniFi Switch Lite 16 PoE provides 16 wired gigabit ports for both Pi servers, the Windows laptop, four operator workstations, a network printer, and room for expansion. This replaces the original 5-port Flex 2.5G which is too small for the full deployment.

DEVICE	ROLE	NOTES
ASUS RT-BE58 Go	Wi-Fi 7 access point / DHCP server / WAN gateway	Dual-band Wi-Fi 7 (802.11be). 2.5G LAN uplink to switch. USB-C 18W power. Handles EMCOMM-NET Wi-Fi, DHCP for 192.168.50.100-200, and optional WAN routing.
UniFi Switch Lite 16 PoE	16-port gigabit managed switch / 8x PoE + 8x non-PoE / 2x 1G SFP uplink	Replaces the 5-port Flex 2.5G. Port layout: 1=ASUS router uplink, 2=FieldCommand Pi, 3=44Net Gateway Pi, 4=Windows laptop, 5=color MFP printer, 6-9=Pi 500 workstations, 10=Satellite dish (optional), 11-16=spare. ~$200. PoE powers future UniFi APs without a separate injector.

Accessories & Cables

ITEM	REQUIRED?	PURPOSE
USB-A drive (32 GB+, labeled FIELDCOMMAND)	Recommended	Auto-backup trigger — insert to back up all runtime data instantly
External USB hard drive 1 TB+ (e.g. LaCie Rugged or LaCie Mobile Drive)	Recommended	Incident archive and full system backup. LaCie Rugged is shock/drop/crush resistant. Label FIELDCOMMAND for auto-backup compatibility.
USB GPS receiver (u-blox or GlobalSat)	Optional	Live position for tactical map and NWS alerts
USB TNC — Digirig Mobile or SignaLink USB	Optional	Required for APRS transmit/receive via Graywolf or YAAC. FieldCommand assigns a stable /dev/tnc0 device name via udev.
Powered USB hub (4- or 7-port, with power supply)	Optional	Recommended if connecting GPS + TNC + backup drive simultaneously. Must be powered — unpowered hubs can cause instability.
USB Printer (e.g. Brother HL-L2350DW, HP LaserJet)	Optional	Shared across EMCOMM-NET via CUPS (installed automatically). Or connect a Wi-Fi printer directly to EMCOMM-NET.

USB Hub & Multi-Device Setup

The Pi 5 has four USB ports. A powered USB hub lets you connect GPS, TNC, backup drive, and other peripherals simultaneously. FieldCommand uses udev rules to assign stable device names regardless of hub port or plug order.

DEVICE	STABLE NAME	HOW ASSIGNED	SERVICE
USB GPS receiver	/dev/gps0	udev rule matches by USB vendor/product ID	gpsd → tactical map
Digirig Mobile TNC	/dev/tnc0 /dev/digirig	udev rule matches by vendor ID + product string "Digirig Mobile". Distinguished from GPS even though both use CP2102 chip.	YAAC / Graywolf APRS
SignaLink USB TNC	/dev/tnc0 /dev/signalink	udev rule matches by Texas Instruments PCM2904 chip. No conflict with GPS.	YAAC / Graywolf APRS
LaCie / USB backup drive (labeled FIELDCOMMAND)	/dev/sdX (auto-mounted)	udev backup rule triggers on label FIELDCOMMAND. Drive letter assigned dynamically.	fieldcommand-backup@.service

Both the Digirig Mobile TNC and GlobalSat BU-353-S4 GPS use the same USB chip (Silicon Labs CP2102, vendor 10c4:ea60). FieldCommand distinguishes them by USB product description string. The GPS rule excludes Digirig; the TNC rule requires the Digirig product string. Verify with: udevadm info -a -n /dev/ttyUSB0 | grep -E "idVendor|idProduct|product"

```
# List all USB serial devices
ls -la /dev/ttyUSB* /dev/ttyACM* 2>/dev/null
# Check the stable symlinks
ls -la /dev/gps0 /dev/tnc0 2>/dev/null
# Confirm GPS is outputting NMEA sentences
sudo cat /dev/gps0
# Use /dev/tnc0 in YAAC → Configure → Ports → Serial TNC
```

Complete Bill of Materials

ITEM	MODEL / SPEC	WHERE TO BUY
— FieldCommand Server (Pi 5 — 16 GB) —		
Raspberry Pi 5 — 16 GB RAM	Raspberry Pi 5 Model B (16 GB) — note: DRAM shortage pricing, ~$305 as of mid-2026 (was $120)	raspberrypi.com · Adafruit · PiShop.us (~$305)
Pironman MAX 5 enclosure	Pironman MAX 5 (tower, dual M.2 NVMe, OLED display, active cooling fans)	52pi.com · Amazon
NVMe SSD × 2 (RAID 1)	1 TB M.2 2280 PCIe Gen 3/4 NVMe (e.g. WD Blue SN580 or Samsung 980) — NAND shortage pricing ~$180 each as of mid-2026 (was ~$75)	Amazon · B&H; · Newegg (~$180 each)
Pi 5 USB-C power supply	Official Raspberry Pi 27W USB-C PD power supply	raspberrypi.com · Adafruit · Amazon
MicroSD card (boot / initial RAID setup)	32 GB Class 10 / A1 microSD (removed after RAID is built)	Amazon
— 44Net Gateway (Pi 5 — 8 GB) —		

ITEM	MODEL / SPEC	WHERE TO BUY
Raspberry Pi 5 — 16 GB RAM	Raspberry Pi 5 Model B (16 GB) — dedicated AMPRNet gateway (matches FieldCommand Pi for spare-parts commonality) — 8 GB variant (~$130) acceptable for gateway-only role	raspberrypi.com · Adafruit · PiShop.us (~$305)
Argon NEO 5 M.2 BRED case	Argon NEO 5 M.2 BRED (compact, passive-cooled, M.2 SATA slot, aluminum) — fits Pi 5	argon40.com · Amazon (~$35)
M.2 SATA SSD (gateway OS + config)	256 GB M.2 2242 or 2280 SATA SSD — OS, WireGuard config, routing tables	Amazon · Newegg (~$25)
Pi 5 USB-C power supply	Official Raspberry Pi 27W USB-C PD power supply — one per Pi	raspberrypi.com · Adafruit · Amazon
— Networking (upgraded from 5-port to 16-port) —		
ASUS RT-BE58 Go travel router	ASUS RT-BE58 Go (Wi-Fi 7, 2.5G LAN uplink, USB-C 18W power)	Amazon · Best Buy · B&H;
UniFi Switch Lite 16 PoE	Ubiquiti USW-Lite-16-PoE (16-port GbE, 8× PoE, 2× SFP uplink, 45W) — replaces Flex 2.5G-5	ui.com · Amazon · B&H; (~$200)
CAT 6 Ethernet cables × 10	1 ft – 6 ft CAT 6 patch cables — router to switch, both Pis, laptop, printer, 4× workstations	Amazon · Monoprice
— Accessories — Backup & Connectivity —		
USB-A drive (32 GB+, labeled FIELDCOMMAND)	32 GB+ USB 3.0 drive — auto-backup trigger when inserted	Amazon
External USB hard drive 1 TB+ (e.g. LaCie Rugged)	1 TB+ USB 3.0/USB-C portable drive — label FIELDCOMMAND for auto-backup	B&H; · Amazon · Best Buy
USB GPS receiver (optional)	GlobalSat BU-353-S4 or u-blox USB GPS puck — provides live position to tactical map and NWS alerts	Amazon
Powered USB hub (optional)	Anker 7-Port USB 3.0 with AC power adapter — must be powered hub, not bus-powered	Amazon · Best Buy
Avery 5371 business card sheets (operator access cards)	Avery 5371 — 10 cards per sheet, laser or inkjet — for printed operator access cards	Office Depot · Amazon · Staples
— Printers — Monochrome Laser (USB via CUPS or direct Wi-Fi) —		
Brother HL-L2350DW (recommended)	Monochrome laser — USB + Wi-Fi — excellent Linux driver — fast, duplex, durable	Best Buy · Amazon (~$130)
HP LaserJet Pro M15w / P1102w	Monochrome laser — USB + Wi-Fi — HPLIP driver auto-installs with CUPS	Best Buy · Amazon (~$150)
— Printers — Color Multifunction Laser (USB, Wi-Fi, or Ethernet) —		

ITEM	MODEL / SPEC	WHERE TO BUY
Brother MFC-L3770CDW	Color laser MFP — print, scan, copy, fax — USB + Wi-Fi + Ethernet — full Linux CUPS support — prints color ICS maps and IAP packages	Amazon · Best Buy (~$400)
HP Color LaserJet Pro MFP M479fdw	Color laser MFP — print, scan, copy, fax — USB + Wi-Fi + Ethernet — HPLIP driver works out of the box with CUPS	Amazon · B&H; (~$500)
Canon imageCLASS MF743Cdw	Color laser MFP — letter + legal — USB + Wi-Fi + Ethernet — Linux CAPT driver — good for permanent EOC installations	Amazon · Office Depot (~$450)
— Printers — Portable / Battery-Powered (field deployments without shore power) —		
Canon PIXMA TR150	Compact inkjet — built-in rechargeable battery — ~200 pages per charge — USB + Wi-Fi — letter size	Best Buy · Amazon (~$200)
HP OfficeJet 200	Portable inkjet — larger paper tray — optional battery accessory — USB + Wi-Fi — good for shelter stations	Best Buy · Amazon (~$180)
— Operator Workstations — Raspberry Pi 500 / 500+ (qty: 4 recommended) —		
Raspberry Pi 500 keyboard computer ×4	Pi 500 standalone — Pi 5 chip, 8 GB RAM, 32 GB A2 microSD, integrated keyboard — one per operator station	raspberrypi.com · Adafruit · Amazon (~$130 each — DRAM shortage pricing)
Raspberry Pi 500+ ×4 (alternative)	Pi 500+ — same as Pi 500 but with Pi 5 latest revision chip and 2× USB3 — specify if 501(c)(3) funding allows premium spec	raspberrypi.com · Adafruit (~$410 each — significantly increased by DRAM shortage; standard Pi 500 recommended)
Raspberry Pi Monitor 15.6" ×4	15.6" Full HD IPS touchscreen — built-in speakers — kickstand — VESA mount — USB-C powered from Pi 500 — one per station	raspberrypi.com · GigaParts · Amazon (~$100 each)
micro-HDMI to HDMI cable ×4	1m micro-HDMI to standard HDMI — Pi 500 to Monitor video connection — one per station	Included in Pi 500 Desktop Kit · Amazon (~$8 each)
USB-C power supply for Pi 500 ×4	Official Raspberry Pi 27W USB-C PD supply — one per station — Pi 500 can also be powered from Monitor USB-C port	raspberrypi.com · Adafruit (~$12 each)
Raspberry Pi 500 Desktop Kit ×4 (bundle option)	Pi 500 + official mouse + micro-HDMI cable bundled — add Monitor separately — simpler ordering than individual parts	raspberrypi.com · Amazon (~$160 each — DRAM shortage pricing)

EMCOMM-NET Coverage Extension — 2× ASUS RT-BE58 Go Mesh Nodes

FieldCommand uses two additional ASUS RT-BE58 Go routers as AiMesh nodes to extend EMCOMM-NET coverage across large EOC buildings, shelters, and outdoor staging areas. Using identical hardware for all three routers (primary + 2 nodes) simplifies deployment, eliminates compatibility issues, and means any unit can serve as the primary router if needed. All three broadcast the same EMCOMM-NET SSID — operators never know they switched nodes.

ITEM	MODEL / SPEC	WHERE TO BUY
ASUS RT-BE58 Go ×2 (mesh nodes)	ASUS RT-BE58 Go — identical to primary router — pairs instantly as AiMesh node with no extra configuration — extends EMCOMM-NET at full Wi-Fi 7 speed — wired backhaul via UniFi switch ports 11 and 12	Amazon · Best Buy · B&H; (~$120-150 each)
USB-C power supply ×2 (for mesh nodes)	USB-C 18W PD adapter — one per node — or USB-C power bank (10,000+ mAh, 18W+) for field use without shore power	Amazon (~$15-25 each)
CAT 6 Ethernet cable ×2 (wired backhaul)	CAT 6 patch cable from UniFi Switch Lite 16 PoE to each node LAN port — wired backhaul is strongly recommended — provides full throughput vs wireless backhaul — length as needed for the venue	Amazon · Monoprice · Home Depot (~$10-20 each)

DEPLOYMENT SCENARIO	RECOMMENDED SETUP
Single room EOC (under 2,500 sq ft)	1× RT-BE58 Go primary only — no extension needed
Multi-room EOC or large shelter (2,500–7,500 sq ft)	1× primary + 1 RT-BE58 Go node — wired backhaul via switch Port 11
Large building or campus (7,500–20,000 sq ft)	1× primary + 2 RT-BE58 Go nodes — wired backhaul via switch Ports 11 and 12
Outdoor SAR staging area	1× primary at command post + 1-2 RT-BE58 Go nodes at field positions on battery

All three ASUS RT-BE58 Go routers broadcast the same EMCOMM-NET SSID and password. Operators and devices roam between them automatically — no reconnection, no password re-entry, no awareness of which router they are connected to. Pairing a new node to the primary takes under 5 minutes on site.

SD Card / Storage Sizing Guide

TIER	WHAT FITS / REQUIREMENT
Minimum (boot only)	8 GB — OS only, RAID boot loader. FieldCommand runs from NVMe RAID.
Recommended	32 GB — OS + swap + logs + temp files during initial RAID setup.
Kiwix Tier 1	~3 GB additional on RAID — WikiMed + Wikipedia Mini + Wikivoyage.
Kiwix Tier 2	~10 GB additional on RAID — adds iFixit, Wikibooks, Wikiversity.
FCC Database	~600 MB on RAID — full US amateur license database (~800K records).

1 Flash OS & Build RAID 1 — Do This First

Step 1: Flash the OS and Build the RAID Array

These two tasks must be completed before anything else. The Raspberry Pi OS must be installed and the RAID 1 array assembled before the FieldCommand installer runs. If you are using a single NVMe drive (no RAID), flash the OS to it directly and skip the RAID build section.

1A Flash the Operating System

OS EDITION	PROS	CONS
Raspberry Pi OS Lite (64-bit) — Recommended for production	Minimal overhead. All RAM available to FieldCommand. Fast boot. Smaller attack surface.	No local browser or GUI. All interaction via SSH or via EMCOMM-NET browser on another device.
Raspberry Pi OS Desktop (64-bit) — Good for setup and troubleshooting	Local browser for testing. GUI for initial Wi-Fi setup and configuration. Easier for new users.	More RAM used by desktop environment. Slightly longer install time.
Ubuntu Server 24.04 LTS — Also supported	LTS kernel. Familiar to Linux admins. Long-term package support.	Some package names differ from Raspberry Pi OS. Slightly more manual setup.

Use **Raspberry Pi Imager** (download from raspberrypi.com/software) to flash the OS to either a microSD card (for single-drive setup) or directly to the NVMe SSD (if not using RAID). During the Imager advanced options, set:

IMAGER OPTION	VALUE / REASON
Hostname	fieldcommand.local — makes the Pi discoverable as fieldcommand.local on the network
Username	fieldcommand — the FieldCommand installer expects this username
Password	Choose a strong password and record it — required for SSH and CUPS admin
Enable SSH	Yes — required for remote access from your laptop during setup
Wi-Fi	Set your home or lab Wi-Fi during initial OS setup. EMCOMM-NET comes later.
Locale / Timezone	US/Central for local operations

1B Build the RAID 1 Array (Pironman MAX 5 — Dual NVMe)

The Pironman MAX 5 enclosure has two M.2 NVMe slots. Configuring them as a RAID 1 mirror means both drives always contain identical data. If one SSD fails, the Pi keeps running on the surviving drive with no data loss and no operator action required. Skip this section entirely if you are using a single NVMe drive.

> ■ Complete the RAID build BEFORE running the FieldCommand installer. The installer must be run on the final storage configuration. If your OS is already running from a single drive, back it up before proceeding.

Physical Installation

SLOT	DRIVE	LINUX DEVICE NAME
M.2 Slot 1 (primary)	1 TB NVMe SSD (e.g. WD Blue SN580)	/dev/nvme0n1
M.2 Slot 2 (mirror)	1 TB NVMe SSD — identical model recommended	/dev/nvme1n1

Boot from a microSD card with Raspberry Pi OS Lite to perform the initial RAID setup. After the array is built and the OS copied to it, the SD card is removed.

Build the Array

```
# Install mdadm (RAID management tool)
sudo apt-get update && sudo apt-get install -y mdadm

# Create the RAID 1 array (--run skips the "all zeroes" check for speed)
sudo mdadm --create --verbose /dev/md0 \
--level=1 --raid-devices=2 \
/dev/nvme0n1 /dev/nvme1n1

# Check build progress (shows resync percentage — takes ~15 min for 1 TB)
cat /proc/mdstat

# Create a filesystem on the array
sudo mkfs.ext4 -L fieldcommand-raid /dev/md0

# Mount and copy the running OS to the RAID array
sudo mkdir -p /mnt/raid
sudo mount /dev/md0 /mnt/raid
sudo apt-get install -y rsync
sudo rsync -axv --progress / /mnt/raid/

# Save the RAID config so it survives reboots
sudo mdadm --detail --scan | sudo tee -a /mnt/raid/etc/mdadm/mdadm.conf
sudo update-initramfs -u -k all

# Update /boot/cmdline.txt to boot from the RAID array instead of the SD
sudo nano /boot/cmdline.txt
# Change: root=PARTUUID=xxxx → root=/dev/md0
# Then reboot — Pi boots from RAID. Verify with:
cat /proc/mdstat
# Expected: md0 : active raid1 nvme0n1[0] nvme1n1[1] [UU]
```

RAID STATUS	MEANING	ACTION REQUIRED
[UU] (both drives active)	Normal — both drives healthy and in sync	None
[U_] or [_U] (one drive failed)	One drive has failed. Pi is still running on the other.	Power down. Replace failed drive. Run: sudo mdadm /dev/md0 --add /dev/nvme1n1. Array rebuilds automatically.
Resync in progress	Array is rebuilding — normal after first setup or drive replacement	Leave running. Takes 15 – 30 minutes. Pi is usable during rebuild.

■ RAID 1 protects against drive failure — it is NOT a backup. Both drives contain identical data, so accidental deletion affects both drives simultaneously. Continue to use the FIELDCOMMAND USB auto-backup for regular data backups.

3. Before You Begin — Prerequisites

Complete these prerequisites before running the installer. The installer requires an internet connection for package downloads, the FCC database, and Kiwix ZIM files. After installation, the server operates fully offline.

3.1 Flash the Operating System

OS EDITION	PROS	CONS
Raspberry Pi OS Lite (64-bit) — Recommended for production	Minimal overhead, faster boot, all RAM available to FieldCommand services, smaller attack surface	No local browser or GUI — all interaction via SSH or EMCOMM-NET browser
Raspberry Pi OS Desktop (64-bit) — Good for setup and troubleshooting	GUI for initial setup, Raspberry Pi Configuration tool, local browser for testing	More RAM used by desktop environment; install takes longer
Ubuntu Server 24.04 LTS — Also supported	LTS kernel, familiar to Linux admins, good for organizations standardized on Ubuntu	Slightly larger memory footprint; some package names differ

3.2 First Boot — Initial Setup

STEP	ACTION
1	On first boot, complete the initial setup wizard: set username to **fieldcommand**, set a strong password, configure locale and keyboard, expand filesystem.
2	If using Desktop edition: complete the initial desktop setup wizard.
3	Enable SSH (Lite only): run `sudo raspi-config → Interface Options → SSH → Enable`

Run a full system update before installing FieldCommand (works the same on both editions — Terminal or SSH):

```
# Update all packages to latest versions
sudo apt-get update && sudo apt-get full-upgrade -y
# Reboot to apply kernel updates
sudo reboot
```

3.3 Verify Disk Space

```
# Check available space — should show 20GB+ free for Tier 2 install
df -h /
# Verify RAM
free -h
```

3.4 Internet Connection

Connect the Pi to the internet via Ethernet before running the installer. Wi-Fi can be used during initial setup but Ethernet is strongly preferred for downloading large files — the FCC database is ~600 MB and Kiwix ZIMs range from 500 MB to 25 GB.

3.5 Operator Workstation — Browser Options

FieldCommand runs entirely in a web browser. Any modern browser on any device connected to EMCOMM-NET works. For the Raspberry Pi 500 / 500+ used as an operator workstation, three browsers are available:

BROWSER	HOW TO INSTALL	NOTES FOR FIELDCOMMAND
Chromium (default)	Pre-installed on Raspberry Pi OS. No action needed.	Best performance for FieldCommand. Discovers CUPS printers automatically via Bonjour. Set http://192.168.50.1 as the startup page under Settings → On startup.
Firefox ESR	sudo apt install firefox-esr	Excellent compatibility with all FieldCommand pages. Good fallback if Chromium behaves unexpectedly. Also discovers network printers. Slightly lower memory footprint on the Pi 500.
GNOME Web (Epiphany)	sudo apt install epiphany-browser	Very lightweight WebKit browser. Good for Pi 500 with many tabs open. Best for dedicated check-in stations where memory is tight.

■ For dedicated FieldCommand operator stations running Pi 500: set Chromium to open http://192.168.50.1 on startup. In Chromium: Settings → On startup → Open a specific page → enter http://192.168.50.1. The dashboard loads automatically every time the Pi 500 boots.

■ Wi-Fi is provided by the ASUS RT-BE58 Go router — the Pi does NOT run hostapd. During installation, connect the Pi to the internet via Ethernet through the UniFi switch and ASUS router. After installation, the Pi is always reachable at 192.168.50.1 via wired Ethernet.

2 Network Hardware Setup — ASUS Router + UniFi Switch

Step 2: Network Hardware Setup — ASUS Router + UniFi Switch

Complete this step before running the FieldCommand installer. The ASUS router must be configured with the correct LAN subnet (192.168.50.x) before FieldCommand is reachable at http://192.168.50.1.

1.1 Physical Wiring

The UniFi Switch Lite 16 PoE is the central wiring hub for the entire EMCOMM-NET deployment. Connect everything to the switch, and the switch connects up to the ASUS router. Use short CAT 6 patch cables (1 ft to 3 ft) to keep the rack or case tidy. The the cellular antenna unit Drum antenna connects directly to the ASUS WAN port via its PoE Ethernet cable — this is completely separate from the switch and does not use a switch port.

	DEVICE	IP / NOTES
et)	the cellular antenna unit Drum (primary cellular WAN)	Primary internet — cellular via your cellular carrier. PoE unit to ASUS WAN port directly. the cellular antenna un subnet — ASUS bridges to EMCOMM-NET.
o-Ethernet)	satellite dish (secondary WAN — automatic failover)	Secondary internet — satellite. satellite Ethernet adapt adapter → ASUS USB port. ASUS switches automatica
	ASUS RT-BE58 Go — LAN 2.5G port	Uplink from router to switch. Router is DHCP server for
	FieldCommand Pi 5 (Pironman MAX 5)	192.168.50.1 — FieldCommand application server. All
	44Net Gateway Pi 5 (Argon NEO 5)	192.168.50.2 — AMPRNet WireGuard gateway. Route EMCOMM-NET devices.
	Windows Laptop (Winlink Express + JS8Call)	192.168.50.3 (recommended static reservation) — or c Wi-Fi.
	Color MFP Network Printer	192.168.50.10 (recommended static reservation) — au via Bonjour.
	Pi 500 Operator Workstations (up to 4)	192.168.50.20 – 192.168.50.23 (static DHCP reservati
	satellite dish Router (if deployed)	Optional second WAN path — connect satellite dish Et as WAN source on ASUS router.
	ASUS RT-BE58 Go — Mesh Node 1	Wired backhaul to first mesh node. Node extends EMC coverage zone.
	ASUS RT-BE58 Go — Mesh Node 2	Wired backhaul to second mesh node. Node extends E coverage zone.
	Spare	Available for additional devices, IP radios, second TNC

DEVICE	POWER SOURCE
ASUS RT-BE58 Go router	USB-C 18W PD adapter — or USB-C power bank for field use
UniFi Switch Lite 16 PoE	Included 45W AC power supply
FieldCommand Pi 5 (Pironman MAX 5)	Official Raspberry Pi 27W USB-C PD supply via Pironman MAX 5 power inlet
44Net Gateway Pi 5 (Argon NEO 5)	Official Raspberry Pi 27W USB-C PD supply directly to Pi USB-C port

DEVICE	POWER SOURCE
ASUS RT-BE58 Go router	USB-C 18W PD adapter — or USB-C power bank for field use
UniFi Switch Flex 2.5G-5	Included USB-C 5V / 1A adapter — or PoE from ASUS router uplink port
Raspberry Pi 5 (in Pironman MAX 5)	Official 27W USB-C PD supply connected to Pironman MAX 5 power inlet

1.2 the cellular antenna unit — Physical Setup and Antenna Mounting

The the cellular antenna unit system consists of an outdoor antenna/modem unit connected by a single PoE Ethernet cable to the ASUS router WAN port. The modem is built into the antenna enclosure — no separate modem box to install. A single cable carries both power to the outdoor unit and data back to the router.

Drum Antenna (Primary — Omnidirectional)

STEP	ACTION
1	**Mount the Drum** on a standard 1.5" to 2" mast, pole, or tripod at the deployment site. Higher is better — roof edge, vehicle roof rack, or portable mast. The Drum is omnidirectional so pointing direction does not matter.
2	**Run the PoE Ethernet cable** from the Drum down to the ASUS router location. The cable carries power to the Drum and data back to the router. Maximum cable length is typically 100 meters (328 ft) for PoE over CAT 6.
3	**Connect the PoE cable** to the **ASUS RT-BE58 Go WAN port**. Do not connect to the switch — this goes directly to the router WAN port.
4	**Power on the the cellular antenna unit system.** The Drum status LED will indicate cellular connection status. Open http://my.insty or http://10.1.1.1 on any device temporarily connected to the the cellular antenna unit Wi-Fi to verify the modem has cellular signal.
5	**Verify the ASUS WAN port shows "Connected"** in the ASUS router admin at http://192.168.50.254 → WAN → Internet Status.

Switchblade Antenna (Backup — Directional)

Deploy the Switchblade when the Drum cannot get adequate signal at a specific site — typically in areas with marginal cellular coverage or significant RF obstruction. The Switchblade folds flat for transport and deploys in under 5 minutes.

STEP	ACTION
1	**Unfold the Switchblade** and mount it on the same mast or a portable tripod.
2	**Open the the cellular antenna unit signal app** on a phone connected to the the cellular antenna unit Wi-Fi. Slowly rotate the Switchblade while watching signal strength in the app. Stop at the direction giving the highest signal — typically facing the nearest the nearest cellular tower.
3	**Unplug the Drum PoE cable** from the ASUS WAN port.
4	**Connect the Switchblade PoE cable** to the same ASUS WAN port.
5	**Verify connection** in the ASUS router admin — WAN should show "Connected" with the the cellular modem IP.

the cellular antenna unit Data Plan — Activation and Standby

STATE	COST	WHEN TO USE	HOW TO CHANGE
Active — Multi-Network	~$79-99/month	During activations, exercises, and training events	Log in to the cellular antenna unit portal — activate plan
Standby mode	$5/month	Between activations — holds your plan and SIM active without full monthly charge	Log in to the cellular antenna unit portal — pause plan (available after first month)

STATE	COST	WHEN TO USE	HOW TO CHANGE
Cancelled	$0	Extended periods of non-use (re-activation required)	Cancel in portal — re-activate and receive new SIM when needed

> For your organization operations, the recommended approach is to keep the plan in Standby mode ($5/month) between activations. This maintains the SIM, the phone number, and the multi-network capability so you can activate at full speed the moment an emergency is declared. The standby cost is negligible for a 501(c)(3) organization and avoids the delay of re-activating a cancelled plan during an actual emergency.

1.2 Configure the ASUS RT-BE58 Go

STEP	ACTION
1	Power on the router. Connect a device to it via Wi-Fi (default SSID on label) or Ethernet. Open http://192.168.50.1 (or the default router IP on the label).
2	Complete the router setup wizard. When it asks for WAN, choose Automatic IP (DHCP) or skip if no internet is connected.
3	Go to **LAN** → **LAN IP**. Set the LAN IP to **192.168.50.1**, subnet **255.255.255.0**. Apply.
4	Go to **LAN** → **DHCP Server**. Set IP pool: **192.168.50.100 – 192.168.50.200**. Apply.
5	Change the ASUS router admin password to something strong. Record it on a piece of paper stored on the FIELDCOMMAND USB drive.
6	Set Manually Assigned IP for the Pi: go to **LAN** → **DHCP Server** → **Manually Assigned IP**. Enter the Pi's MAC address → assign IP **192.168.50.1**. (Or set the Pi's static IP manually in Step 4.)
7	Set the Wi-Fi SSID: go to **Wireless** → **General**. Set both the 2.4 GHz and 5 GHz SSIDs to **EMCOMM-NET**. Set a strong WPA3/WPA2 password. Apply.

The ASUS RT-BE58 Go supports dual WAN with automatic failover. the cellular modem is the primary WAN source. satellite dish is the secondary — the ASUS switches automatically if cellular drops. When both are unavailable, FieldCommand continues operating on all local features without interruption — net logging, ICS platform, APRS, and all tools remain fully functional.

ONNECTION	SETUP	ACTIVATE
hernet port (PoE from modem)	PoE Ethernet cable from the cellular antenna unit outdoor unit to ASUS WAN port. Set WAN → Automatic IP (DHCP). the cellular antenna unit modem serves DHCP.	Always —
AN Ethernet port (replaces Drum cable)	Unplug Drum PoE cable. Plug Switchblade PoE cable into same ASUS WAN port. Aim Switchblade toward nearest tower using the cellular antenna unit signal app.	Manual —
SB WAN port (via Ethernet adapter)	satellite Ethernet adapter → USB-to-Ethernet adapter → ASUS USB WAN port. Set ASUS WAN failover: primary = WAN port, secondary = USB port. Automatic failover when cellular WAN drops.	Automatic
hernet port (replaces the cellular antenna unit)	Plug site Ethernet directly into ASUS WAN port. Set WAN → Automatic IP (DHCP). Use when site has reliable wired internet.	Manual —
SB WAN port	Enable hotspot on phone. Plug USB into ASUS router USB port. Set WAN → USB Tethering.	Manual —

ONNECTION	SETUP	ACTIVATE
ireless WAN	ASUS Web UI → Operation Mode → WISP. Select venue Wi-Fi and enter password.	Manual —

1.3 Configure Dual WAN Failover (the cellular antenna unit Primary + satellite dish Secondary)

With the cellular antenna unit connected to the WAN port and satellite dish connected via USB, configure the ASUS router to switch automatically between them. This takes about 3 minutes and requires no changes to FieldCommand software.

STEP	ACTION
1	Open the ASUS router admin: **http://192.168.50.254**
2	Go to **WAN** → **Dual WAN**.
3	Set **Enable Dual WAN** to ON.
4	Set **Primary WAN** to **WAN** (the physical WAN port — the cellular antenna unit).
5	Set **Secondary WAN** to **USB** (the USB port — satellite dish via adapter).
6	Set **Dual WAN mode** to **Failover mode**.
7	Under **Heartbeat server**, set the primary check target to **1.1.1.1** (Cloudflare) and secondary to **8.8.8.8** (Google). These are the IPs the router pings to confirm WAN is up.
8	Set **Failback to primary WAN when primary WAN is recovered** to ON. This switches back to the cellular antenna unit automatically when cellular returns.
9	Click **Apply**.
10	Test failover: unplug the the cellular antenna unit PoE cable from the WAN port. Within 30-60 seconds, the ASUS should switch to satellite dish USB. Check WAN status in the router admin — it should show USB WAN active. Reconnect the cellular antenna unit — the router should failback to WAN port within 60 seconds.

WAN STATE	WHAT EMCOMM-NET DEVICES SEE	FIELDCOMMAND IMPACT
the cellular antenna unit UP (primary)	Full internet — cellular 5G/LTE via your cellular carrier	NWS weather alerts live, APRS-IS active, FCC DB refresh available, Pat Winlink via internet
the cellular antenna unit DOWN → satellite dish UP (failover)	Full internet via satellite — slightly higher latency (~20-40ms typical)	All internet features remain active. CGNAT on satellite dish means no inbound connections possible.
Both WAN sources DOWN	No internet — EMCOMM-NET still fully operational	All local FieldCommand features work normally. NWS alerts, APRS-IS, FCC refresh paused until WAN returns.

1.4 AiMesh Mesh Node Setup — Extending EMCOMM-NET Coverage

The standard FieldCommand deployment includes three ASUS RT-BE58 Go routers: one primary (connected to the cellular antenna unit WAN and the UniFi switch) and two mesh nodes that extend EMCOMM-NET to additional rooms, floors, or outdoor areas. All three use the same hardware, the same SSID, and the same password. Devices roam between them automatically.

> Connect the two mesh nodes via wired Ethernet backhaul (switch Ports 11 and 12) rather than wireless backhaul whenever possible. Wired backhaul provides full Wi-Fi 7 throughput on both upstream and downstream — wireless backhaul cuts bandwidth roughly in half because the node uses its radios for both the backhaul link and the client connections simultaneously.

Repeat these steps for each of the two mesh nodes:

STEP	ACTION

1	**Connect the node Ethernet** — run a CAT 6 cable from UniFi Switch Port 11 (first node) or Port 12 (second node) to the LAN port of the mesh node router. This is the wired backhaul — it carries all traffic between the node and the primary.
2	**Power on the node** — connect the USB-C power supply to the mesh node. For initial pairing, position the node within 30 feet of the primary router. After pairing it can be moved to its final location.
3	**Open AiMesh on the primary router** — go to **http://192.168.50.254** → **AiMesh** → **AiMesh Node**. The new node appears in the device list within 1-2 minutes.
4	**Click Connect** on the new node. The primary router pushes the EMCOMM-NET SSID, password, and configuration to the node automatically. This takes 1-3 minutes.
5	**Move the node to its final position** — mount or place it in the coverage area it needs to serve. The wired backhaul cable connects back to the UniFi switch. Test coverage with a phone on EMCOMM-NET at the furthest point in the area.
6	**Verify in AiMesh dashboard** — on the primary router, AiMesh → AiMesh Node should show both nodes as Connected with backhaul type Ethernet. Green status on both nodes confirms the mesh is fully operational.

NODE	SWITCH PORT	RECOMMENDED PLACEMENT
Primary RT-BE58 Go (router)	Uplink to ASUS 2.5G LAN port	Central to the deployment — EOC command post or entrance area
Mesh Node 1 (first RT-BE58 Go)	UniFi Switch Port 11	Secondary room, opposite wing, or upper floor
Mesh Node 2 (second RT-BE58 Go)	UniFi Switch Port 12	Third coverage zone — outdoor staging, parking, or far building wing

3 Download & Run the FieldCommand Installer

Step 3: Download & Run the FieldCommand Installer

Method A — Direct download to the Pi (requires internet on Pi)

Lite: SSH into the Pi and run these commands. **Desktop:** open a Terminal window (taskbar → Terminal) and run the same commands.

```
# Create working directory
mkdir -p ~/fieldcommand-install && cd ~/fieldcommand-install
# Download the FieldCommand package from GitHub or your storage location
wget -O fieldcommand-v1.zip
https://github.com/KE4CON/CrossPlatformAPRS/releases/latest/download/fieldcommand-v1.zip
# Unzip the package
unzip fieldcommand-v1.zip
cd fieldcommand
```

Method B — Copy from your computer using SCP (Lite/headless)

```
# Run this on YOUR COMPUTER, not the Pi
scp fieldcommand-v1.zip fieldcommand@[pi-ip-address]:~/
# Then SSH into the Pi and unzip
ssh fieldcommand@[pi-ip-address]
mkdir -p ~/fieldcommand-install
mv fieldcommand-v1.zip ~/fieldcommand-install/
cd ~/fieldcommand-install && unzip fieldcommand-v1.zip
cd fieldcommand
```

Method B2 — Download directly on the Pi Desktop (Desktop edition)

OPTION	HOW
Browser download	Open Chromium on the Pi. Go to the GitHub releases page or download URL. Download fieldcommand-v1.zip — it saves to ~/Downloads/ automatically.
USB drive	Copy fieldcommand-v1.zip to a USB drive. Insert the USB into the Pi. Open File Manager, drag the zip to your home folder.

```
# If downloaded to ~/Downloads/:
mkdir -p ~/fieldcommand-install
cp ~/Downloads/fieldcommand-v1.zip ~/fieldcommand-install/
# Unzip (same for all methods):
cd ~/fieldcommand-install && unzip fieldcommand-v1.zip
cd fieldcommand
```

Method C — USB drive transfer (fully offline)

Lite: commands below via SSH. **Desktop:** use File Manager to copy the zip from USB to your home folder, then open a Terminal for the unzip step.

```
# Copy fieldcommand-v1.zip to a USB drive on your computer
# Insert USB drive into the Pi, then:
sudo mount /dev/sda1 /mnt # or check lsblk for your device
cp /mnt/fieldcommand-v1.zip ~/fieldcommand-install/
cd ~/fieldcommand-install && unzip fieldcommand-v1.zip
cd fieldcommand
```

Launch the Installer

Lite: run via SSH. **Desktop:** open a Terminal window from the taskbar or Accessories → Terminal. The installer runs identically on both.

```
cd ~/fieldcommand-install/fieldcommand
sudo bash scripts/install.sh
```

4 Installer Configuration Options

Step 4: Installer Configuration Options

PROMPT	DEFAULT	DESCRIPTION
Station callsign	W8ABC	Your amateur callsign. Patched into all HTML pages.
Station latitude	42.3153	Decimal degrees. Used for APRS station marker, NWS alerts, propagation, distance calc.
Station longitude	-88.4473	Decimal degrees (negative = West). Default 0.0/0.0 — set to your deployment coordinates.
Wi-Fi AP SSID	EMCOMM-NET	The Wi-Fi network name that field devices connect to.
Wi-Fi AP password	fieldcommand2026	WPA2 password for EMCOMM-NET. Change this for operational security.
Server IP address	192.168.50.1	The Pi's static IP on EMCOMM-NET. Devices browse to http://192.168.50.1/
Download FCC database	N	~600 MB download of the FCC amateur license database. Recommended: Y
Kiwix tier	1	Tier 1: WikiMed + Wikipedia Mini + Wikivoyage (~2.5 GB). Tier 2 adds iFixit (~5 GB more).

What the Installer Does Automatically

- Installs system packages: Python 3, nginx, kiwix-tools, rsync, gpsd, ufw, mdadm, java
- Creates the **fieldcommand** system user and directory structure at /opt/fieldcommand/
- Creates a Python virtual environment and installs Flask, flask-cors, requests, gpsd-py3
- Deploys all 30 HTML pages to /opt/fieldcommand/html/ and patches your callsign and coordinates into each file
- Installs all 14 systemd service and timer units and enables them at boot
- Configures nginx with the correct web root (/opt/fieldcommand/html/) and proxy rules for all API ports
- **Does NOT configure a Wi-Fi access point** — Wi-Fi is handled by the ASUS RT-BE58 Go router. The Pi connects as a wired client only.
- Configures the Pi static IP (192.168.50.1/24) on the Ethernet interface (eth0) via NetworkManager
- Configures the firewall (ufw) — opens all required ports
- Downloads and installs YAAC (Java APRS client) and Graywolf to /opt/yaac/ and /opt/graywolf/
- Installs the USB backup udev rule (plug in a USB drive labeled FIELDCOMMAND to trigger auto-backup)
- Downloads and installs Pat Winlink (browser-based Winlink backup client, port 8090)
- Runs the Kiwix ZIM downloader for the selected content tier
- Downloads the FEMA ICS forms PDFs to /opt/fieldcommand/data/ics_forms/ (22 forms)
- Optionally downloads and builds the FCC amateur license database (~600 MB)
- Starts all services and verifies they are running

> ■ The ASUS RT-BE58 Go router must be configured before running the installer — see Step 1. The Pi does NOT run hostapd or dnsmasq. All Wi-Fi, DHCP, and SSID management is handled by the ASUS router.

5 Raspberry Pi 5 — Static IP Configuration

Step 5: Raspberry Pi 5 — Static IP Configuration

The Pi must always be reachable at 192.168.50.1. If the installer already set this via a DHCP reservation in the ASUS router, verify with `ip addr show eth0` and skip to Step 5 if 192.168.50.1/24 is shown.

Path A — Raspberry Pi OS Lite (SSH)

```
# SSH into the Pi (use the IP shown in your router DHCP table)
ssh fieldcommand@[current-pi-ip]
# Find the Ethernet connection name
nmcli con show
# Usually: "Wired connection 1" or "Preconfigured Connection 1"
# Set static IP
sudo nmcli con mod "Wired connection 1" ipv4.addresses 192.168.50.1/24 ipv4.method manual
ipv4.method manual
# Apply the change
sudo nmcli con up "Wired connection 1"
# Verify — should show inet 192.168.50.1/24
ip addr show eth0
```

Path B — Raspberry Pi OS Desktop (GUI)

STEP	ACTION
1	Click the **network icon** in the taskbar (top-right corner).
2	Click **Advanced Options** → **Edit Connections**.
3	Select your **Wired connection** and click the gear/edit icon.
4	Click the **IPv4 Settings** tab.
5	Change **Method** from "Automatic (DHCP)" to **Manual**.
6	Click **Add** and enter: Address: 192.168.50.1 · Netmask: 255.255.255.0 · Gateway: (leave blank)
7	Click **Save**, then close the Network Connections window.
8	Disconnect and reconnect the Ethernet cable, or reboot the Pi.
9	Open the **Terminal** and verify: `ip addr show eth0` — should show **inet 192.168.50.1/24**.
10	Open **Chromium** on the Pi and go to http://192.168.50.1 to confirm FieldCommand loads locally.

■ Alternatively, open a Terminal on the desktop and use the same nmcli commands shown in Path A. The terminal method works identically on the Desktop edition.

Path C — raspi-config (if available)

```
# Open raspi-config:
sudo raspi-config
# Navigate to: System Options → Network → (set static IP if option is present)
# NOTE: Static IP configuration via raspi-config was removed in Raspberry Pi OS Bookworm.
# If the option is not visible, use Path A (nmcli) or Path B (GUI Network Connections).
```

4b

RAID 1 NVMe Setup — Pironman MAX 5

7b. RAID 1 NVMe Setup — Pironman MAX 5

The Pironman MAX 5 enclosure has two M.2 NVMe slots. Configure them as a RAID 1 mirror so both drives contain identical data. If one drive fails, the system continues running on the surviving drive with no data loss and no operator action required.

> Complete this step before running the FieldCommand installer. The RAID array must be assembled and the OS installed on it before FieldCommand is deployed. If your Pi OS is already on a single drive, back it up first.

Physical Installation

STEP	ACTION
1	Install both 1 TB NVMe SSDs into the Pironman MAX 5 M.2 slots. Slot 1 (primary) → /dev/nvme0n1 · Slot 2 (mirror) → /dev/nvme1n1
2	Boot from a Raspberry Pi OS microSD card for the initial RAID setup. After RAID is built and OS is installed, the SD card is removed.

Build the RAID 1 Array

```
# Install mdadm (RAID management tool)
sudo apt-get install -y mdadm
# Create RAID 1 array from the two NVMe drives
sudo mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 \
/dev/nvme0n1 /dev/nvme1n1
# Confirm array is building (will show [=>..............] resync progress)
cat /proc/mdstat
# Create a filesystem on the RAID array
sudo mkfs.ext4 -L fieldcommand-raid /dev/md0
# Mount it
sudo mkdir -p /mnt/raid
sudo mount /dev/md0 /mnt/raid
# Copy running OS to RAID array
sudo apt-get install -y rsync
sudo rsync -axv --progress / /mnt/raid/
# Save RAID configuration so it persists across reboots
sudo mdadm --detail --scan | sudo tee -a /mnt/raid/etc/mdadm/mdadm.conf
```

RAID Health Reference

SITUATION	WHAT TO DO
Both drives show [UU]	Normal healthy state. No action needed.
One drive fails — [_U] or [U_]	1. Power down. 2. Remove failed drive. 3. Insert new 1 TB NVMe. 4. Power on. 5. Run: sudo mdadm /dev/md0 --add /dev/nvme1n1. 6. Array rebuilds automatically (~30 min).
Check RAID health anytime	cat /proc/mdstat or sudo mdadm --detail /dev/md0

RAID 1 protects against drive failure — it is NOT a backup. Both drives contain identical data, so accidental deletion or corruption affects both drives simultaneously. Continue to use the USB auto-backup (insert FIELDCOMMAND USB drive) for regular data backups.

5 Kiwix Offline Library Setup

8. Step 5: Kiwix Offline Library Setup

Kiwix provides an offline web library served on port 8081. All devices on EMCOMM-NET can browse it at http://192.168.50.1:8081. Content is packaged as ZIM files.

TIER	NAME	SIZE	DESCRIPTION
Essential	WikiMed — Medical Encyclopedia	~471 MB	Offline medical encyclopedia: symptoms, treatments, drugs, procedures. Critical for mass-casualty events.
Essential	Wikipedia (Mini)	~1.2 GB	Compact English Wikipedia — essential facts, geography, science.
Essential	Wikivoyage	~820 MB	Emergency shelters, evacuation routes, local resources and travel logistics.
Extended	Wikibooks — How-To & Field Manuals	~4.2 GB	First aid, survival, ham radio, electronics, cooking, field skills.
Extended	iFixit — Equipment Repair Manuals	~3.1 GB	Step-by-step repair guides for electronics, tools, vehicles.
Extended	Wikipedia (Full English)	~22 GB	Complete English Wikipedia. Large download — requires 30 GB+ free.

```
# Check what is installed and service status
sudo bash /opt/fieldcommand/scripts/kiwix_setup.sh --status

# Add Tier 2 content (resumes interrupted downloads automatically)
sudo bash /opt/fieldcommand/scripts/kiwix_setup.sh --tier 2

# Service management
sudo systemctl status kiwix
sudo systemctl restart kiwix # required after adding new ZIM files
```

■ ZIM downloads are resumable. If a download is interrupted by power loss or network drop, simply re-run kiwix_setup.sh with the same --tier flag. curl resumes from where it stopped — no need to start over.

Additional Kiwix Content — Install Any ZIM File

Kiwix.org (kiwix.org/en/content/) publishes hundreds of ZIM archives beyond the two built-in tiers. Any ZIM file can be added to FieldCommand. The following table lists the most useful additions for emergency communications and public safety operations:

CONTENT	APPROX SIZE	BEST FOR
OpenStreetMap (Wikimedia Maps)	~500 MB	Offline detailed maps for any region. Download your county or state for SAR and shelter mapping.
Stack Overflow (programming Q&A;)	~3.5 GB	Technical troubleshooting reference. Useful for net managers debugging FieldCommand.
Project Gutenberg (public domain books)	~65 GB (full) / ~1 GB (best-of)	Field reference library. Includes FCC regulations, ARRL handbook excerpts, first aid guides.

CONTENT	APPROX SIZE	BEST FOR
Khan Academy (education)	~30 GB	Training resource for shelter and EOC staff during extended activations.
Red Cross First Aid (WikiMed companion)	~200 MB	Practical first aid procedures in plain language — supplements WikiMed for non-medical staff.
Wiktionary (dictionary)	~520 MB	Reference for unfamiliar ICS terminology and acronyms.
Amateur Radio Wiki (if available)	Variable	Band plans, antenna theory, Winlink and APRS procedures.

How to Install Any ZIM File

```
# Step 1 — Find and download the ZIM from a device with internet
# Browse to: https://library.kiwix.org
# Filter by language (English) and category.
# Download the .zim file to your laptop or phone.

# Step 2 — Transfer the ZIM to the Pi
# Option A: USB drive (copy to USB, insert in Pi)
sudo mount /dev/sda1 /mnt
cp /mnt/my_content.zim /opt/kiwix/zim/

# Option B: SCP from your laptop over EMCOMM-NET
scp my_content.zim fieldcommand@192.168.50.1:/opt/kiwix/zim/

# Step 3 — Register the new ZIM with the Kiwix library index
sudo kiwix-manage /opt/kiwix/library.xml add /opt/kiwix/zim/my_content.zim

# Step 4 — Restart Kiwix to load the new content
sudo systemctl restart kiwix

# Step 5 — Verify it appears at http://192.168.50.1:8081
# The new content appears in the Kiwix home page search bar.

# To list all installed ZIM files:
ls -lh /opt/kiwix/zim/

# To remove a ZIM file:
sudo rm /opt/kiwix/zim/my_content.zim
sudo systemctl restart kiwix
```

■ ZIM files can be large. Check available disk space before downloading: df -h /opt/kiwix/ The RAID array provides 2× 1 TB = ~900 GB usable after OS and FieldCommand. Tier 1 uses ~2.5 GB. Tier 2 uses ~10 GB. There is ample space for additional content.

6 FCC Amateur Database

9. Step 6: FCC Amateur Database

The FCC database gives FieldCommand instant offline access to the entire US amateur radio license database — approximately 800,000 active licensees. When a net control operator types a callsign into the check-in field, FieldCommand looks up the licensee name, license class, and address automatically in under a millisecond, with no internet connection required. This is one of the most-used features during ARES/RACES net operations.

The installer offers to download and build the database during Step 4 (answer Y). If you skipped it during installation, or if the database needs to be rebuilt, run the commands below. Building takes approximately 5 to 10 minutes on a Pi 5 and requires an active internet connection.

```
# If you selected "y" during install, it already ran.
# To build or rebuild manually:
sudo -u fieldcommand \
/opt/fieldcommand/venv/bin/python \
/opt/fieldcommand/python/build_fcc_db.py
# Automatic weekly refresh timer:
sudo systemctl status fcc-refresh.timer
# Force immediate refresh:
sudo systemctl start fcc-refresh.service
```

7 APRS Setup — Graywolf + YAAC

10. Step 7: APRS Setup (Graywolf + YAAC)

FieldCommand integrates two APRS client applications: Graywolf TNC (accessible on port 8080) and YAAC — Yet Another APRS Client (port 8082). Both run as background services and feed live station data to the Tactical APRS Map on the dashboard. The map merges data from both sources, deduplicates by callsign, and maintains a live WebSocket feed to all connected browsers simultaneously.

APRS is entirely optional. FieldCommand runs fully without it — you simply will not see live APRS stations on the tactical map. If you are not planning to use a TNC or radio interface, skip this step entirely and proceed to Step 8 (Printer Setup).

■ The FieldCommand installer attempts to download and install both YAAC and Graywolf automatically. If internet was available during installation, they should already be present. Skip to Section 9.3 (YAAC port configuration) if the installer succeeded.

9.1 Verify Automatic Installation

```
# Check if Java is installed
java -version
# Check if YAAC was installed
ls -lh /opt/yaac/YAAC.jar
# Check if Graywolf was installed
ls -lh /opt/graywolf/graywolf.jar
# Check service status
sudo systemctl status yaac
sudo systemctl status graywolf
```

9.2 Manual Install — If Automatic Install Failed

```
# Step 1 — Install Java runtime (required by both clients)
sudo apt install -y default-jre
# Step 2 — Install YAAC
sudo mkdir -p /opt/yaac
cd /tmp && wget http://www.ka2ddo.org/ka2ddo/YAAC.zip
sudo unzip YAAC.zip "*.jar" -d /opt/yaac/
sudo mv /opt/yaac/YAAC*.jar /opt/yaac/YAAC.jar 2>/dev/null || true
sudo chown -R fieldcommand:fieldcommand /opt/yaac
# Step 3 — Install Graywolf
sudo mkdir -p /opt/graywolf
sudo wget -O /opt/graywolf/graywolf.jar \
https://github.com/vk2tds/graywolf/releases/latest/download/graywolf.jar
sudo chown -R fieldcommand:fieldcommand /opt/graywolf
# Step 4 — Enable services
sudo systemctl enable --now yaac
sudo systemctl enable --now graywolf
```

9.3 YAAC Port Configuration (Required — First Run)

STEP	ACTION
1	Run YAAC once on a desktop/monitor: `java -jar /opt/yaac/YAAC.jar`
2	Go to **File** → **Configure** → **Web Server** tab.
3	Set **Port: 8082** · Check **Enable REST API** · Check **Enable WebSocket** · Click Save.

4	Close YAAC. The yaac.service will use these settings when running headlessly.

9.4 Connect a TNC or Radio Interface

INTERFACE TYPE	PORT TYPE IN YAAC	SETTINGS
USB TNC (Digirig, SignaLink, etc.)	Serial TNC	/dev/tnc0, baud rate 9600 or 1200
KISS TNC over TCP (Direwolf)	KISS TNC (TCP)	Host: localhost, Port: 8001
AGW packet engine (Direwolf)	AGW	Host: localhost, Port: 8000
APRS-IS internet (if WAN connected)	APRS-IS	Server: noam.aprs2.net:14580, Filter: r/42.31/-88.44/50

9.5 Verify APRS is Working

```
# Test YAAC REST API (should return JSON list of heard stations)
curl http://localhost:8082/api/stations
# Test Graywolf REST API
curl http://localhost:8080/api/stations
# Check service logs if no stations appear
journalctl -u yaac -n 30
journalctl -u graywolf -n 30
```

8 Printer Setup — Three Connection Options

11. Step 8: Printer Setup

FieldCommand has print buttons on 17 pages. All printing uses the browser standard window.print() function, which sends the job to whatever printer the operator's browser can reach. Three printer connection methods are supported:

Option A — Client Device Prints to Its Own Printer (Simplest)

Each operator's laptop or tablet prints to its own locally connected printer. No Pi configuration needed. Zero setup required — works immediately.

> ■ This is the simplest option and requires zero configuration on the Pi. The only requirement is that the operator's device has a printer configured.

Option B — USB Printer Shared via Pi (CUPS)

A USB printer is connected directly to the Pi. CUPS shares it across EMCOMM-NET. Installed automatically by the FieldCommand installer. CUPS admin UI at http://192.168.50.1:631.

B.2 Add the USB Printer via CUPS Web Admin

STEP	ACTION
1	Plug the USB printer into one of the Pi's USB ports (or powered USB hub).
2	On any device on EMCOMM-NET, open a browser and go to: **http://192.168.50.1:631**
3	Click **Administration** → **Add Printer**.
4	If prompted for credentials, enter the Pi username and password (e.g. fieldcommand / your password).
5	Your USB printer appears in the **Local Printers** list. Select it and click **Continue**.
6	Enter a Name (e.g. FieldCommand-Printer), Description, and Location. Check **Share This Printer**. Click Continue.
7	Select the correct printer driver. Click **Add Printer**.
8	Set default options (paper size: Letter) and click **Set Default Options**.
9	Click **Print Test Page** to confirm the printer is working.

B.3 Add the Shared Printer on Operator Devices

DEVICE TYPE	HOW TO ADD THE SHARED PRINTER
Windows laptop (Winlink / JS8Call station)	Settings → Bluetooth & devices → Printers & scanners → Add a printer or scanner. Windows discovers the CUPS shared printer automatically on EMCOMM-NET via Bonjour. Select FieldCommand-Printer when it appears and click Add device.
Mac / macOS	System Settings → Printers & Scanners → click + (Add Printer). Click the Default tab — the shared printer appears via Bonjour discovery. Select it and click Add.
iPad / iPhone (AirPrint)	No setup required. CUPS + Avahi advertises the printer as AirPrint-compatible. Open any FieldCommand page → tap the Share icon → tap Print → select the printer. Works on any iOS device on EMCOMM-NET.
Android tablet / phone	Install the free CUPS Print app from Google Play. Open the app → Add Printer → enter IP 192.168.50.1, Port 631, Protocol IPP. The printer appears in the Android print dialog for all apps.

DEVICE TYPE	HOW TO ADD THE SHARED PRINTER
Chromebook	Settings → Advanced → Printing → Printers → Add Printer. Enter: Name = FieldCommand-Printer, Address = 192.168.50.1, Protocol = IPP, Queue = /printers/FieldCommand-Printer. Click Add.
Raspberry Pi 500 / 500+ (operator workstation)	Chromium browser (pre-installed): the CUPS printer appears automatically in the print dialog. No additional setup needed — Chromium on Pi OS discovers CUPS printers via Bonjour. For Firefox ESR on the Pi 500: install via sudo apt install firefox-esr then open Print → More Settings → select the printer from the destination list. Alternatively, run: sudo usermod -aG lpadmin fieldcommand to give the Pi 500 user full access to the CUPS admin at http://192.168.50.1:631.

B.4 Recommended Printers for Field Use

Monochrome Laser (recommended for most EOC deployments)

PRINTER	CONNECTION	WHY IT WORKS WELL
Brother HL-L2350DW	USB or Wi-Fi	Excellent Linux driver support out of the box. Fast, compact, duplex, durable. Best all-round choice for field EOC use. Under $130.
HP LaserJet Pro M15w / P1102w	USB or Wi-Fi	Full Linux support via HP HPLIP driver (installed automatically with CUPS). Fast, reliable, very low cost per page. Under $150.

Color Multifunction Laser (ICS maps, colored forms, and chart printing)

PRINTER	CONNECTION	WHY IT WORKS WELL
Brother MFC-L3770CDW	USB, Wi-Fi, or Ethernet	Color laser all-in-one with duplex, scan, copy, fax. Full Linux support via Brother driver. Prints color ICS maps and resource boards clearly. Under $400.
HP Color LaserJet Pro MFP M479fdw	USB, Wi-Fi, or Ethernet	Color laser multifunction — print, scan, copy, fax. HPLIP driver works with CUPS out of the box. Excellent for IAP packages with color-coded sections. Under $500.
Canon imageCLASS MF743Cdw	USB, Wi-Fi, or Ethernet	Color laser multifunction with excellent Linux CUPS driver (CAPT driver). Handles letter and legal paper. Good choice for permanent EOC installations. Under $450.

Portable / Battery-Powered (mobile and field deployments without shore power)

PRINTER	CONNECTION	WHY IT WORKS WELL
Canon PIXMA TR150	USB or Wi-Fi (battery built-in)	Compact inkjet with rechargeable battery. Prints ~200 pages per charge. Letter size. Wi-Fi connects directly to EMCOMM-NET. Under $200.
HP OfficeJet 200	USB or Wi-Fi (battery optional)	Portable inkjet with larger paper tray than TR150. Optional battery accessory. Good for shelter check-in stations and mobile command posts. Under $180.

Option C — Network Printer on EMCOMM-NET (Simplest Shared Setup)

A Wi-Fi or Ethernet printer is connected directly to the ASUS router or UniFi switch — no Pi involvement required at all. All devices on EMCOMM-NET can discover and print to it automatically via Bonjour / mDNS. This works with any network-capable printer including color multifunction laser printers that have built-in Wi-Fi or Ethernet. This is the simplest option for permanent EOC installations where a color MFP is already on site.

Option C works with any network-capable printer — including all three color multifunction laser printers listed in the BOM. These are the best choice for a permanent EOC installation because they connect directly to EMCOMM-NET with no Pi configuration, provide color printing for ICS maps and IAP packages, and are discoverable by every device on scene automatically.

CONNECTION METHOD	COMPATIBLE PRINTERS	HOW TO SET UP
Wi-Fi (wireless)	Brother MFC-L3770CDW / HP Color LaserJet M479fdw / Canon MF743Cdw / Canon PIXMA TR150 / HP OfficeJet 200	Use the printer control panel to join EMCOMM-NET. Enter the Wi-Fi password when prompted. The printer receives a 192.168.50.x address from the ASUS router automatically. All EMCOMM-NET devices discover it via Bonjour — no driver installation needed.
Ethernet (wired)	Brother MFC-L3770CDW / HP Color LaserJet M479fdw / Canon MF743Cdw	Run a CAT 6 cable from the printer Ethernet port to any spare UniFi Switch port (Ports 3 through 5). The printer receives an IP automatically. Set a static DHCP reservation in the ASUS router so the IP never changes between activations.

Reserve a Static IP for the Printer (Recommended)

```
# On the ASUS router web UI (http://192.168.50.1):
# LAN → DHCP Server → Manually Assigned IP
# Enter the printer MAC address → assign a fixed IP, e.g.: 192.168.50.10
# Reboot the printer to pick up the reservation
# Confirm: ping 192.168.50.10
```

Printer Option Comparison

	Option A — Client Printer	Option B — USB via Pi / CUPS	Option C — Network Printer
Pi configuration needed	None	CUPS (auto-installed)	None
One printer for all operators	Each device needs own printer	Shared across EMCOMM-NET	Shared across EMCOMM-NET
Works without internet	Yes	Yes	Yes
iOS / AirPrint support	Requires AirPrint printer	Yes via CUPS + Avahi	Yes if AirPrint printer
Battery printer support	Yes	Yes (USB connected)	Yes (Wi-Fi connected)
Best for	Single operator or tablet with own printer	Any USB printer, shared to all devices	Wi-Fi/Ethernet printer already on site

■ For most your callsign field activations, Option B (USB printer via CUPS) is recommended. A Brother HL-L2350DW or HP LaserJet plugged into the Pi USB hub covers all ICS forms, net logs, and cheat sheets needed for a typical activation.

9 Pat Winlink — Verify & Configure

12. Step 9: Pat Winlink — Verify & Configure

Pat is an open-source Winlink client that runs directly on the Pi and is accessible from any browser on EMCOMM-NET at http://192.168.50.1:8090. The FieldCommand installer downloads and configures Pat automatically, so this step is primarily about verifying it is running and adding your Winlink account password.

Pat serves as the backup Winlink path when the Windows laptop is unavailable, or for VHF Packet operations when a TNC is connected directly to the Pi. your organization primary Winlink operations use Winlink Express on the Windows laptop (Step 10). Pat is the fallback.

```
# Check pat.service status
sudo systemctl status pat
# If not running:
sudo systemctl start pat && sudo systemctl enable pat
# Confirm port 8090 is listening
ss -tlnp | grep 8090
# Add your Winlink password:
sudo nano /opt/fieldcommand/.config/pat/config.json
# Verify / update these fields:
# "mycall": "your callsign" ← set by installer
# "secure_login_password": "xxxxx" ← add your Winlink password
# "http_addr": "0.0.0.0:8090" ← must be 0.0.0.0 for EMCOMM-NET access
```

10 Windows Laptop — Winlink Express + JS8Call

12. Step 10: Windows Laptop Setup

The Windows laptop is the primary HF digital station. It runs Winlink Express and JS8Call with the IC-7300 connected via a single USB cable. Connect the laptop to EMCOMM-NET (Wi-Fi) or the UniFi switch Port 3 (wired Ethernet).

9.1 IC-7300 One-Time Radio Setup

IC-7300 MENU PATH	SETTING
SET → Connectors → CI-V Baud Rate	115200
SET → Connectors → CI-V Transceive	ON
SET → Connectors → USB Send/Keying → USB Send	RTS
SET → Connectors → MOD Input → USB MOD Level	40–50%
COMP button (speech compression)	OFF
Mode for digital operation	USB-D

9.2 Install Winlink Express + VARA HF

STEP	ACTION
1	Download Winlink Express from: **winlink.org/client-software**. Run installer, accept defaults.
2	On first launch: enter your callsign, Winlink password, and grid square (your county).
3	Settings → Radio Setup: Radio=IC-7300, Control Port=(check Device Manager), Baud Rate=115200, PTT via CAT=checked.
4	Settings → Sound Card: Input=USB Audio CODEC (IC-7300), Output=USB Audio CODEC (IC-7300).
5	Download and install VARA HF from: winlink.org → VARA → VARA HF Modem. Set same audio devices in VARA HF Settings.
6	Test: Open a VARA HF session in Winlink Express → connect to any Winlink RMS gateway on 40m to confirm audio and CAT control work.

9.3 Install JS8Call

STEP	ACTION
1	Download JS8Call from: **js8call.com** → Windows installer. Run installer, accept defaults.
2	Launch JS8Call. Open File → Settings (F2). General tab: My Call=, My Grid=.
3	Audio tab: Input=USB Audio CODEC (IC-7300), Output=USB Audio CODEC (IC-7300).
4	Radio tab: Rig=IC-7300, PTT Method=CAT, Serial Port=(same COM port as Winlink), Baud Rate=115200.
5	Reporting tab (CRITICAL): TCP Server Hostname=**0.0.0.0** (MUST change from 127.0.0.1). Enable TCP Server API=checked. TCP Server Port=**2442**. Accept TCP Requests=checked.

6	Click OK. Restart JS8Call for API settings to take effect.
7	Set VFO to 7.078 MHz (JS8 40m calling frequency). Mode → Normal.

9.4 Windows Firewall — Allow JS8Call Port 2442

```
1. Search "Windows Defender Firewall" → Advanced Settings
2. Inbound Rules → New Rule → Port → TCP → Specific port: 2442 → Allow → All profiles
3. Name the rule: JS8Call API
```

9.5 Find the Windows Laptop IP on EMCOMM-NET

```
# Run in Windows Command Prompt:
ipconfig
# Note the IPv4 Address for the EMCOMM-NET adapter (e.g. 192.168.50.105)
# Set a DHCP reservation (recommended — gives laptop a fixed IP):
# ASUS Web UI → LAN → DHCP Server → Manually Assigned IP
# Enter laptop MAC address → assign 192.168.50.2

# Configure the FieldCommand dashboard JS8Call card:
# 1. Connect to EMCOMM-NET, open http://192.168.50.1
# 2. Find the JS8Call card (purple) in Amateur Radio section
# 3. Tap/click the card
# 4. Enter the Windows laptop IP (e.g. 192.168.50.105)
# 5. Card saves IP and opens http://192.168.50.105:2442
```

SWITCHING TO...	PROCEDURE
Winlink Express (from JS8Call)	Click Disconnect in JS8Call or close it. Open VARA HF — it automatically reclaims the USB audio device. Open Winlink Express and connect normally.
JS8Call (from Winlink)	Finish any active Winlink transmission. Close VARA HF. Minimize Winlink Express. Open JS8Call and click Connect — it reclaims the COM port and audio.
Both simultaneously (two radios)	Connect a second HF radio via a separate USB audio and CAT interface. Configure JS8Call to use that radio's COM port and audio device. Both run at the same time without conflicts.

10b SCS PACTOR Modem Setup (Optional — High-Performance HF)

Step 10b: SCS PACTOR Modem Setup

PACTOR is a high-performance HF digital mode used extensively in Winlink operations. It provides significantly better throughput than VARA HF on marginal or noisy HF paths, and is the only mode that supports PACTOR 4 compression on Winlink. SCS (Special Communications Systems) manufactures the only PACTOR-licensed TNCs available. A PACTOR TNC connects between the IC-7300 and the Windows laptop. This step is optional — Winlink via VARA HF works well for most activations.

Supported SCS PACTOR Modems

MODEL	PACTOR LEVEL	INTERFACE	NOTES
SCS Tracker DSP TNC	PACTOR 1 / 2 / 3	USB (COM port)	Entry-level PACTOR TNC. Good for most Winlink operations. Also supports PACKET and ROBUST PR. ~$250.
SCS PTC-IIIusb	PACTOR 1 / 2 / 3	USB (COM port)	Full-featured controller with front-panel display. Supports PACTOR, PACKET, AMTOR, RTTY. ~$650.
SCS P4dragon DR-7400	PACTOR 1 / 2 / 3 / 4	USB (COM port)	Top-of-line PACTOR 4 TNC. Maximum throughput on HF. Required for PACTOR 4 sessions. ~$1,100. Recommended for permanent EOC and EMCOMM stations.

Physical Connection

FROM	TO	CABLE / NOTE
SCS TNC — AFSK audio output	IC-7300 — ACC jack audio input or 3.5mm mic input	SCS-supplied audio cable (included with TNC)
SCS TNC — AFSK audio input	IC-7300 — ACC jack audio output or 3.5mm speaker output	SCS-supplied audio cable (included with TNC)
SCS TNC — PTT output	IC-7300 — remote jack (ACC or SEND)	SCS-supplied PTT cable (check your TNC model)
SCS TNC — USB port	Windows laptop — any USB-A port	USB-A to USB-B or USB-A to micro-USB (included with TNC)

If you are also using VARA HF via the IC-7300 USB audio: configure the PACTOR TNC to use the IC-7300 ACC jack for audio, not the USB sound card. This allows VARA (USB audio path) and PACTOR (ACC audio path) to coexist on the same radio. Never transmit both modes simultaneously.

Configure Winlink Express for PACTOR

STEP	ACTION
1	In Winlink Express, go to **Settings** → **Radio Setup**.

2	Click **Add** to create a new radio entry.
3	Set **Template** to: **SCS PTC-IIx** (works for all SCS PACTOR TNCs including the Tracker and P4dragon).
4	Set **Control Port** to the COM port assigned to your SCS TNC. Check Windows Device Manager → Ports (COM & LPT) to find the correct number. The TNC appears as "USB Serial Port (COM X)".
5	Set **Baud Rate** to **115200**.
6	Leave **PTT via CAT** unchecked — the TNC handles PTT directly via the ACC cable.
7	Click **OK** to save the radio profile.

Open a PACTOR Winlink Session

STEP	ACTION
1	In Winlink Express, click the session type dropdown and select **Open Session** → **PACTOR Winlink**.
2	The session window opens. The TNC initializes and you will see the SCS banner text in the session log.
3	Set your frequency on the IC-7300 to a known Winlink RMS gateway frequency for your region. Common 40m PACTOR frequencies: 7.038 MHz, 7.040 MHz, 7.103 MHz.
4	Click **Start**. Winlink Express scans for and connects to the nearest RMS gateway.
5	The PACTOR handshake completes automatically — the session connects at the highest common level (PACTOR 1 through 4) that both stations support.
6	Send and receive messages as normal. PACTOR 4 typically completes a full ICS-213 exchange in under 2 minutes on a good path.

Verify the TNC is Responding

```
# Method: use PuTTY or Windows Terminal to open a direct serial session
# Open PuTTY → Connection type: Serial → COM port: (your TNC port)
# Speed: 115200 → click Open

# Type the following TNC command:
cmd
# The TNC responds with its command prompt: cmd:

# Useful TNC verification commands:
ver # shows firmware version and TNC model
status # shows current operating status
mycall YOURCALL # replace YOURCALL with your FCC callsign

# If the TNC does not respond:
# 1. Confirm the correct COM port in Device Manager
# 2. Confirm no other application has the COM port open
# 3. Try baud rate 9600 (some TNC models default lower)
```

PACTOR levels 3 and 4 require a separate Winlink license fee beyond the standard Winlink account. PACTOR 1 and 2 are available to all licensed amateur radio operators at no additional cost. The SCS TNC also supports AX.25 Packet, which Pat Winlink on the Pi can use via a KISS TCP connection (typically for VHF packet via a separate TNC or via the Digirig). This provides a VHF backup to the HF PACTOR path.

11 AMPRNet / 44Net Gateway Setup

Step 11: AMPRNet / 44Net Gateway Setup

The following files are part of FieldCommand and work together to provide integrated 44Net gateway support across both Pis:

FILE	RUNS ON	PURPOSE
scripts/setup_44net.sh	Gateway Pi (192.168.50.2)	Installer for the gateway Pi. Sets up WireGuard, static IP, IP forwarding, status service, and Desktop autostart. Run once on the gateway Pi.
python/amprgate_status.py	Gateway Pi (192.168.50.2)	Flask status service on port 9000. Serves the gateway dashboard UI and /api/status JSON endpoint. Installed by setup_44net.sh.
systemd/amprgate-status.service	Gateway Pi (192.168.50.2)	Systemd unit that keeps amprgate_status.py running. Installed by setup_44net.sh. Starts after wg-quick@ampr0.
python/amprgate_poll.py	FieldCommand Pi (192.168.50.1)	Polls the gateway Pi every 30 seconds and writes status to /opt/fieldcommand/data/amprgate_status.json for the dashboard.
systemd/amprgate-poll.service	FieldCommand Pi (192.168.50.1)	Systemd unit for the poll service. Installed automatically by the FieldCommand install.sh.
html/amprgate.html	FieldCommand Pi (192.168.50.1)	Dashboard page at http://192.168.50.1/amprgate.html. Shows tunnel state, AMPRNet address, traffic stats, and tunnel controls.

AMPRNet (Amateur Packet Radio Network) is the global amateur radio IP network using the reserved 44.0.0.0/8 address block assigned by IANA to ARRL. It provides licensed amateur radio operators with globally routable IP addresses reachable over RF links or via encrypted internet tunnels to the AMPRNet gateway. Adding a dedicated 44Net gateway Pi to FieldCommand gives every device on EMCOMM-NET full access to the AMPRNet — Winlink gateways, APRS-IS servers, Mesh networking nodes, and other amateur stations worldwide that are reachable on 44.x.x.x addresses.

> The 44Net Gateway Pi is completely separate from the FieldCommand server Pi. It runs WireGuard, ip routing, and a status page only. If the gateway goes down or needs reconfiguring, FieldCommand continues operating normally. This isolation was an intentional design choice — keep routing infrastructure separate from the communications server.

Part A — Get Your AMPRNet Allocation (Do This First — Allow 2-4 Weeks)

Before configuring any software, you need an AMPRNet IP address block assigned to your callsign. This is a one-time registration process managed by the AMPRNet administrators at ampr.org. The block you receive will be something like 44.x.x.x/29 (6 usable addresses) or 44.x.x.x/28 (14 usable addresses) depending on what is available in your region.

STEP	ACTION
1	**Create an account at portal.ampr.org** — go to https://portal.ampr.org and click Register. You must use your callsign your callsign as your username. Verify your license via the FCC ULS lookup that the portal performs automatically.
2	**Request a subnet allocation** — after logging in, click Subnets → Request Subnet. Select your region (Illinois — typically 44.24.x.x or 44.172.x.x range). Request a /29 block for a small deployment or a /28 if you anticipate growth. In the justification field, explain the your organization EMCOMM-NET deployment and intended use.
3	**Coordinate with your regional AMPRNet administrator** — Illinois AMPRNet administration is coordinated through ARRL. You may be contacted by the regional admin for verification. Response times vary from days to several weeks. Be patient.

4	**Receive your allocation** — you will get an email with your assigned 44.x.x.x/29 block. The portal will also show your allocation under Subnets → My Subnets. Record this allocation — you will need it in Part B.
5	**Note your WireGuard endpoint credentials** — the portal provides a WireGuard configuration for your specific subnet. Download or copy this — it contains your private key, allowed IPs, and the amprgw.ampr.org endpoint address.

ARRL membership is not required for an AMPRNet allocation, but a valid FCC amateur radio license (any class) is required. The your club or personal callsign is valid for this purpose. The allocation is tied to the callsign and is permanent as long as the license remains active.

Part B — Set Up the 44Net Gateway Pi

Once you have your AMPRNet allocation, the following steps configure the dedicated gateway Pi running in the Argon NEO 5 case. This Pi should be freshly flashed with Raspberry Pi OS Lite (64-bit). It does not need the FieldCommand installer — it runs WireGuard and routing only.

B.1 Initial OS Setup

Flash Raspberry Pi OS Desktop (64-bit) to the M.2 SATA SSD using Raspberry Pi Imager. Desktop is used on the gateway Pi so any team member can check tunnel status from the local keyboard and monitor without SSH. Chromium opens the status page automatically on login. During Imager advanced options set:

OPTION	VALUE
Hostname	amprgate.local — discoverable as amprgate.local on EMCOMM-NET
Username	amprgate — used by the gateway setup script
Password	Choose a strong password and record it with the FieldCommand credentials
Enable SSH	Yes — useful for remote diagnostics if the monitor is not available
Wi-Fi	Your home or lab Wi-Fi for initial package downloads — switched to wired EMCOMM-NET after setup
Locale	US/Central — matches the FieldCommand Pi for consistent log timestamps

B.2 Assign Static IP and Install WireGuard

```
# SSH into the gateway Pi
ssh amprgate@[current-pi-ip]

# Update packages
sudo apt-get update && sudo apt-get full-upgrade -y

# Install WireGuard
sudo apt-get install -y wireguard wireguard-tools

# Set static IP on eth0
sudo nmcli con mod "Wired connection 1", ipv4.addresses 192.168.50.2/24, ipv4.gateway 192.168.50.254,
ipv4.dns "8.8.8.8 8.8.4.4", ipv4.method manual
sudo nmcli con up "Wired connection 1"

# Verify
ip addr show eth0
# Expected: inet 192.168.50.2/24
```

B.3 Configure the WireGuard Tunnel

The AMPRNet portal provides a WireGuard configuration specific to your subnet. The following is the general structure — replace the placeholder values with the actual values from your portal allocation.

```
# Create the WireGuard config directory
sudo mkdir -p /etc/wireguard
sudo chmod 700 /etc/wireguard

# Create the tunnel config file
sudo nano /etc/wireguard/ampr0.conf

# Paste the following — replace ALL values in [brackets] with your portal values:
[Interface]
PrivateKey = [your-private-key-from-portal]
Address = [your-44.x.x.x/29-address]
ListenPort = 51820
PostUp = ip route add 44.0.0.0/8 dev ampr0
PostUp = iptables -A FORWARD -i ampr0 -j ACCEPT
PostUp = iptables -A FORWARD -o ampr0 -j ACCEPT
PostUp = iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
PostDown = ip route del 44.0.0.0/8 dev ampr0
PostDown = iptables -D FORWARD -i ampr0 -j ACCEPT
PostDown = iptables -D FORWARD -o ampr0 -j ACCEPT
PostDown = iptables -t nat -D POSTROUTING -o eth0 -j MASQUERADE

[Peer]
PublicKey = [amprnet-gateway-public-key-from-portal]
Endpoint = amprgw.ampr.org:51820
AllowedIPs = 44.0.0.0/8
PersistentKeepalive = 25
```

B.4 Enable IP Forwarding and Start the Tunnel

```
# Enable IP forwarding (required for routing between EMCOMM-NET and AMPRNet)
echo "net.ipv4.ip_forward = 1" | sudo tee -a /etc/sysctl.conf
sudo sysctl -p

# Start the WireGuard tunnel
sudo wg-quick up ampr0

# Enable at boot
sudo systemctl enable wg-quick@ampr0

# Check tunnel status
sudo wg show ampr0
# Expected: shows latest handshake, transfer stats, allowed IPs

# Test reachability into AMPRNet
ping 44.0.0.1 -c 4
# Expected: replies from 44.0.0.1 (AMPRNet gateway)
```

B.5 Access Control — Who Can Reach What

FieldCommand restricts access to the 44Net gateway on two levels. This is both a security measure and a Part 97 compliance measure — only licensed amateur radio operators should be able to view or control the AMPRNet gateway.

PORT	ACCESSIBLE FROM	AUTH REQUIRED	PROVIDES
Port 9000 (0.0.0.0)	Any EMCOMM-NET device (192.168.50.x)	None for /api/status (machine poll). Valid FCC callsign for web UI.	Read-only status JSON. Callsign-authenticated web dashboard.

PORT	ACCESSIBLE FROM	AUTH REQUIRED	PROVIDES
Port 9001 (127.0.0.1)	Gateway Pi keyboard only (localhost — not reachable from network)	Valid callsign session from port 9000 login. Physical presence required.	Tunnel control: up / down / restart.

Tunnel controls (up/down/restart) are intentionally inaccessible from the network. An operator who wants to restart the WireGuard tunnel must sit at the gateway Pi keyboard, open Chromium, log in with their FCC callsign, and use the local control interface. This prevents accidental or unauthorized tunnel disruption from operator laptops or phones. The FieldCommand Pi polling service reads /api/status on port 9000 without authentication — this is a read-only machine-to-machine call and is intentional.

B.5 Advertise the 44Net Route to EMCOMM-NET Devices

For other devices on EMCOMM-NET to use the 44Net gateway automatically, the ASUS router needs to push a static route to all DHCP clients. This tells every device: "to reach anything in 44.0.0.0/8, use 192.168.50.2".

STEP	ACTION
1	Open the ASUS router admin at **http://192.168.50.254**.
2	Go to **LAN** → **Route**.
3	Click **Add** and enter: Network/Host IP = **44.0.0.0**, Netmask = **255.0.0.0**, Gateway = **192.168.50.2**, Interface = **LAN**.
4	Click **Apply**.
5	On any EMCOMM-NET device, verify the route was received:

```
# On any EMCOMM-NET device (Pi 500, Windows laptop, etc.)
ip route show | grep 44
# Expected: 44.0.0.0/8 via 192.168.50.2 dev eth0

# Windows equivalent (Command Prompt)
route print | find "44."
# Expected: 44.0.0.0 0.0.0.0 192.168.50.2 ...

# Test from the FieldCommand Pi
ping 44.0.0.1 -c 4
# If reachable, FieldCommand can now connect to AMPRNet resources
```

B.6 Gateway Status Page

Install a lightweight status page on the 44Net Pi so operators can verify the tunnel state from any EMCOMM-NET browser without SSH access. This is a small Flask application that displays tunnel status, connected peers, and recent handshake times.

```
# Install Python and Flask on the 44Net Pi
sudo apt-get install -y python3 python3-pip python3-venv
python3 -m venv /opt/amprgate/venv
/opt/amprgate/venv/bin/pip install flask

# Create the status app
sudo mkdir -p /opt/amprgate
sudo nano /opt/amprgate/status.py

# Paste the following into status.py:
# ---
from flask import Flask, jsonify
import subprocess, json
app = Flask(__name__)

@app.route("/")
def index():
return open("/opt/amprgate/status.html").read()

@app.route("/api/status")
def status():
wg = subprocess.run(["sudo","wg","show","ampr0","dump"],
capture_output=True, text=True).stdout
lines = [l.split(" ") for l in wg.strip().split(" ") if l]
return jsonify({"tunnel":"up" if lines else "down","peers":len(lines)-1})

if __name__ == "__main__":
app.run(host="0.0.0.0", port=9000)
# ---

# Create systemd service
sudo nano /etc/systemd/system/amprgate-status.service
# Paste:
[Unit]
Description=AMPRNet Gateway Status Page
After=network.target wg-quick@ampr0.service

[Service]
User=amprgate
ExecStart=/opt/amprgate/venv/bin/python /opt/amprgate/status.py
Restart=always

[Install]
WantedBy=multi-user.target

sudo systemctl enable --now amprgate-status
# Status page now at: http://192.168.50.2:9000
```

Part C — What You Can Do on AMPRNet

Once the gateway is operational and the route is advertised to EMCOMM-NET, every device on EMCOMM-NET has full access to the AMPRNet. Here are the most useful applications for your organization operations:

	HOW IT USES 44NET	ACCESS FROM EMCOMM-NET
	Some Winlink RMS gateways have 44.x.x.x addresses. Pat Winlink on the FieldCommand Pi can connect to them directly over the AMPRNet tunnel rather than the commercial internet — useful when only amateur radio paths are available.	Pat Winlink at http://192.168.50.1:8090 Add AMPRNet RMS
	The APRS-IS network has servers reachable on 44.x.x.x. Graywolf and YAAC can connect to APRS-IS via the AMPRNet path, keeping APRS traffic within the amateur radio network.	Configure APRS-IS server address in Graywolf or YAAC se
	If another your organization station runs a second FieldCommand system with its own 44Net gateway, the two systems can share net data and resource boards over AMPRNet without routing traffic through the commercial internet.	Future capability — requires second FieldCommand deploy
	Some VoIP amateur radio nodes are reachable on 44.x.x.x addresses, providing RF-to-AMPRNet linking without commercial internet dependency.	Direct connection to 44.x.x.x node addresses
	Any amateur station worldwide with a 44.x.x.x address (whether via RF packet or WireGuard tunnel) is directly reachable from any EMCOMM-NET device for data exchange, file transfer, or status checking.	Direct TCP/IP connections to 44.x.x.x addresses

> AMPRNet does not carry voice traffic and is not a replacement for your radio links. It is a data network for IP-based amateur radio applications. All traffic on AMPRNet is subject to Part 97 rules — no encryption of content, no commercial traffic, licensed operator identification required. WireGuard encryption of the tunnel itself is permitted as it is the transport, not the content.

Part D — Verification Checklist

CHECK	COMMAND / METHOD	EXPECTED RESULT
AMPRNet portal shows allocation	Login to portal.ampr.org → Subnets → My Subnets	Your 44.x.x.x/29 block shows Status: Active
WireGuard tunnel is up on gateway Pi	SSH to 192.168.50.2 then: sudo wg show ampr0	Shows latest handshake within last 2 minutes
AMPRNet gateway reachable from gateway Pi	ping 44.0.0.1 -c 4	Replies received — round trip time typically 10-50 ms
EMCOMM-NET devices have 44 route	ip route show | grep 44 (on any EMCOMM-NET device)	44.0.0.0/8 via 192.168.50.2
AMPRNet reachable from FieldCommand Pi	ping 44.0.0.1 -c 4 (on FieldCommand Pi)	Replies received
Status page accessible	Open http://192.168.50.2:9000 in any browser on EMCOMM-NET	Shows callsign login page — enter a valid FCC callsign to proceed
Callsign login works	Enter your callsign at http://192.168.50.2:9000 login page	FCC database validates callsign — dashboard loads showing tunnel status
Access log recording	SSH to gateway Pi: tail /var/log/amprgate-access.log	Shows LOGIN-OK entry with your callsign, IP, and license class

CHECK	COMMAND / METHOD	EXPECTED RESULT
Tunnel control blocked from network	From a laptop on EMCOMM-NET: curl http://192.168.50.2:9001	Connection refused — port 9001 is localhost-only
AMPRNet reachable from Pi 500 workstation	Open http://192.168.50.2:9000 in Chromium on Pi 500	Status page loads and shows tunnel UP

12 First Boot Verification — Full System

Step 12: First Boot Verification

After the installer completes and the Pi reboots, run through this verification checklist before handing off the system. Every service should show "active" and the dashboard should load cleanly from another device on EMCOMM-NET. If anything is not running, the troubleshooting section at the end of this guide covers the most common issues and their fixes.

```
# Check all FieldCommand services
for svc in fcc-lookup health-monitor deadmans ics-platform kiwix nginx pat; do
echo "$svc: $(systemctl is-active $svc)"
done

# Test the dashboard from the Pi itself
curl -s http://localhost/ | head -5
# Test API endpoints
curl http://localhost:5050/health # FCC API health check
curl http://localhost:5051/health # System health monitor
# Verify Pi has the correct static IP
ip addr show eth0
# Should show: inet 192.168.50.1/24
```

14. Network Architecture Reference

Full Network Topology

The EMCOMM-NET deployment uses the ASUS RT-BE58 Go as the Wi-Fi access point and DHCP server. The the cellular antenna unit Drum provides primary cellular WAN (your cellular carrier) via a PoE Ethernet cable directly to the ASUS WAN port. satellite dish provides automatic secondary WAN failover via the ASUS USB WAN port. The UniFi Switch Lite 16 PoE distributes wired connections to both Pi servers, all operator workstations, the network printer, and additional field devices. The 44Net Gateway Pi routes the 44.0.0.0/8 AMPRNet address block for all EMCOMM-NET devices via a WireGuard tunnel to amprgw.ampr.org.

DEVICE	IP ADDRESS	ROLE	CONNECTION
ASUS RT-BE58 Go	192.168.50.254 (LAN gateway)	Wi-Fi 7 AP + DHCP server + WAN gateway	WAN: site Ethernet, satellite dish, or USB tether (optional). EMCOMM-NET SSID on 2.4 GHz and 5 GHz.
UniFi Switch Lite 16 PoE	N/A (Layer 2 switch)	Wired distribution hub — 16 ports	Port 1 uplink to ASUS router. Ports 2-16 to all wired devices.
FieldCommand Pi 5 (Pironman MAX 5)	192.168.50.1 (static)	FieldCommand application server All EmComm tools and services	Wired GbE via UniFi Switch Port 2. Does not run Wi-Fi.
44Net Gateway Pi 5 (Argon NEO 5)	192.168.50.2 (static)	AMPRNet / 44Net WireGuard gateway Routes 44.0.0.0/8 for EMCOMM-NET	Wired GbE via UniFi Switch Port 3. WireGuard tunnel to amprgw.ampr.org.
Windows Laptop	192.168.50.3 (DHCP reservation)	Winlink Express + JS8Call IC-7300 HF station	Wired via Port 4 or Wi-Fi (EMCOMM-NET).
Color MFP Printer	192.168.50.10 (DHCP reservation)	Network printer shared to all devices	Wired via Port 5 or Wi-Fi (EMCOMM-NET).
Pi 500 Workstations x4	192.168.50.20-23 (DHCP reservations)	Operator browser stations	Wired via Ports 6-9 or Wi-Fi (EMCOMM-NET).
ASUS RT-BE58 Go Mesh Node 1	Managed by AiMesh	EMCOMM-NET extension (same SSID, seamless roaming)	Wired backhaul via UniFi Port 11. Extends coverage to secondary zone.
ASUS RT-BE58 Go Mesh Node 2	Managed by AiMesh	EMCOMM-NET extension (same SSID, seamless roaming)	Wired backhaul via UniFi Port 12. Extends coverage to third zone.
Field Devices (phones, tablets, laptops)	192.168.50.100-200 (DHCP)	Additional operator browsers	Wi-Fi — EMCOMM-NET — roam between primary and mesh nodes automatically.

■ Recommended: set the ASUS router LAN IP to 192.168.50.254 and the Pi static IP to 192.168.50.1. This keeps the FieldCommand URL clean and the router admin clearly separate at http://192.168.50.254.

13. Web Dashboard Reference

FieldCommand IMS serves 49 web pages from the Pi. All are accessible at http://192.168.50.1/[page]. Pages marked WAN require internet for live data but load and display cached data when offline.

PAGE	URL	NOTES
Dashboard (All-Hazards ICS)	/	Main hub — 3 modes, WAN status bar, tool cards. Default entry point.
Organization Setup	/setup.html	Configure org name, callsign, grid square, time zone, SSID
Preflight Check	/preflight.html	GO/CAUTION/NO-GO readiness. Run before every activation.
Incident Management	/incident.html	Create/manage incidents, operational periods, incident type
Incident Archive/Reset	/incident_mgmt.html	Archive to USB · Restore · Hard delete · Beta/Scenario Reset
Event Templates	/event_templates.html	6 built-in templates (Shelter/SAR/HazMat/Weather/Gathering/Exercise)
General Info / ICS-201	/general_info.html	Initial incident briefing — situation summary, initial actions
ICS Platform — Overview	/ics/index.html	Five-section ICS structure navigator
ICS — Command Section	/ics/command.html	ICS-202 objectives, ICS-208 safety, staff roster, position checklists
ICS — Operations Section	/ics/operations.html	T-card board, resource status, assignment, T-card personnel roster
ICS — Planning Section	/ics/planning.html	IAP assembly, ICS-205/206/209/211/221 forms, Planning P cycle
ICS — Logistics Section	/ics/logistics.html	ICS-205 comms (COML), ICS-206 medical (MEDL), channel library
ICS — Finance/Admin	/ics/finance.html	FEMA PA cost tracking, ICS-214 log, expenditure summary
Planning P Cycle Guide	/ics/planningp.html	15-phase IAP cycle with agenda, required forms, attendees per phase
T-Card Resource Board	/resources.html	Add/status/assign resources. T-card personnel roster per card.
NIMS Resource Types	/resource_types.html	Pre-loaded NIMS types + custom types. Available in T-card board.
GPS Resource Map	/resource_map.html	Live GPS pins per T-card — 3 placement methods, auto-refresh 30s
Starcom Resource Map	/resmap.html	Public safety unit positioning map with zone drawing tools
ICS-213 General Message	/ics213.html	Formal message with digital signature, print, Winlink export
ICS-214 Activity Log	/ics214.html	Unit activity log — export hours directly to FEMA labor costs
ICS-309 Comms Log	/ics309.html	Manual communications log with incident archiving
ICS-204A Briefing Sheet	/briefing_204a.html	Assignment briefing for branch/division supervisors
ICS Form Suite	/ics-form.html	Full ICS form set: ICS-202 through ICS-221 with digital signatures
IAP Assembly	/iap.html	IAP tracker — all forms for active incident with completion status
IAP Compile (PDF)	/iap_compile.html	One-click IAP PDF: title page + section dividers + all forms
FEMA PA Cost Documentation	/fema_costs.html	Force Account Labor, Equipment, Materials, Contracts, PW export

PAGE	URL	NOTES
FEMA Equipment Rates	/fema_rates.html	44 FEMA equipment categories — edit rates, track rate year
Cost Dashboard	/cost_dashboard.html	Live cost totals by category, burn rate, budget tracker, by period
Personnel Accountability	/accountability.html	PAR board — ICS-211, T-card personnel, Unaccounted, Cross-Reference
ICS-211 Check-In	/checkin.html	Manual ICS-211 check-in form with active list and Check Out buttons
QR/Barcode Scan Check-In	/scan_checkin.html	Camera scan or manual ID fallback. BarcodeDetector API, no library.
Member Roster	/roster.html	Directory with certs, equipment, QR code generation, CSV import/export
Amateur Net Control Logger	/netcontrol.html	Multi-net, FCC autofill, check-in/out, traffic, ICS-309, observer link
Public Safety Net Logger	/starcom.html	Radio ID check-in, Starcom/P25 nets, ICS-309 export
Net Observer Mode	/observer.html	Read-only net view, auto-refresh 15s. Share URL with EOC.
Dead Man's Switch	/deadmans.html	Audible alert if net goes silent beyond configured interval
Tactical APRS Map	/tactical.html	Leaflet — RF APRS always, APRS-IS when WAN up, SARTopo overlay
SARTopo GeoJSON Import	/sartopo_import.html	Import planning overlays from SARTopo — persists on tactical map
NEXRAD Radar	/radar.html	WAN REQUIRED. Animated NEXRAD loop — play/pause/scrub/speed/palette
NWS/HF Propagation	/propagation.html	WAN: SFI, K-index, A-index from NOAA SWPC. Band-by-band conditions.
FCC Callsign Lookup	/callsign.html	Offline — 800K+ US licensees in local SQLite. Instant results.
NTS Radiogram Generator	/nts.html	ARRL-format traffic with auto word count. Print or log.
Winlink Form Import	/winlink-import.html	Import Winlink Express XML forms into incident archive
AMPRNet Gateway Status	/amprgate.html	44Net WireGuard tunnel status, callsign auth, Part 97 access log
WAN Status Detail	/wan-status.html	Both WAN sources, detection method, last check, event log
WAN Settings	/wan_settings.html	Configure WAN sources: role, type, detection method, enable/disable
Hospital Directory	/hospitals.html	Trauma level, helipad, phone. Feeds ICS-206 medical plan.
Facilities Directory	/facilities.html	ICP, staging, shelters, supply depots. Links to T-card assignments.
Repeater Database	/repeaters.html	Filter by band/mode/CTCSS. Feeds ICS-205 channel picker.
Channel Library	/channel_library.html	Pre-loaded interop channels (NIFOG/VTAC). Add custom agency channels.
Grid Square Calculator	/grid.html	Grid ↔ lat/lon · distance and bearing between two grids

PAGE	URL	NOTES
Radio Cheat Sheets	/cheatsheets.html	Phonetic alphabet, Q-codes, RST scale, ICS structure, HF frequencies
ICS Position Checklists	/position_checklists.html	NIMS activation checklists for all ICS positions
Meeting Scheduler	/meetings.html	Schedule operational briefings — time, location, attendees
Print Center	/printcenter.html	Print any ICS form or document to any CUPS printer on EMCOMM-NET
Reference Library	/refs.html	Kiwix offline — WikiMed, Wikipedia, iFixit. No internet needed.
Health Monitor	:5051/health	Live CPU/memory/disk/temp, all service status, GPS, internet
Pat Winlink Client	:8090	Browser-based Winlink client (Packet, VARA HF, Telnet)
Kiwix Library	:8080/	Offline Wikipedia, WikiMed, iFixit, Wikivoyage

14. Service & Port Reference

All services run on the FieldCommand Pi (192.168.50.1) unless noted. The AMPRNet gateway runs on a separate Pi (192.168.50.2).

PORT	SERVICE	SYSTEMD UNIT	DESCRIPTION
80	nginx	nginx.service	Web frontend — all 49 HTML pages, static assets
5050	FCC Lookup Server	fieldcommand-fcc.service	FCC callsign lookup, org config, incidents, net logs
5051	ICS Platform	fieldcommand-main.service	ICS forms, T-cards, check-ins, FEMA costs, GPS, PAR
8080	Kiwix Library	kiwix.service	Offline reference — WikiMed, Wikipedia, iFixit
8090	Pat Winlink	pat.service	Winlink email-over-radio — Packet, VARA HF, VARA FM
631	CUPS Printer	cups.service	USB printer shared to all EMCOMM-NET devices
2442	JS8Call	(laptop app)	HF digital keyboard messaging — web API on Windows laptop
9000	AMPRNet Status	amprgate-status.service	AMPRNet gateway read-only status — public on EMCOMM-NET
9001	AMPRNet Control	amprgate-status.service	Tunnel control — localhost only on gateway Pi

Background Services

SYSTEMD UNIT	FUNCTION
fieldcommand-main.service	ICS platform server (port 5051) — T-cards, FEMA, PAR, GPS, check-in
fieldcommand-fcc.service	FCC lookup + org config server (port 5050)
wan-monitor.service	WAN source monitoring and automatic failover every 30 seconds
wan-monitor.timer	Timer that keeps wan-monitor running on schedule

SYSTEMD UNIT	FUNCTION
amprgate-status.service	44Net gateway status API (ports 9000/9001 on gateway Pi)
amprgate-poll.service	44Net WireGuard tunnel keepalive and reconnect on gateway Pi
backup.service / backup.timer	Nightly SQLite backup to labelled USB drive
kiwix.service	Kiwix offline reference library server (port 8080)
pat.service	Pat Winlink radio email client (port 8090)
nginx.service	Web frontend — serves all HTML pages and static assets (port 80)
cups.service	CUPS print server — shares USB printer to EMCOMM-NET

15. New Feature Configuration

FieldCommand IMS v1.0 includes the following features beyond the base ICS platform. Most require no additional setup — they are available immediately after installation. Items that require initial configuration are noted below.

15.1 Personnel Accountability (PAR)

The Personnel Accountability page (/accountability.html) is available immediately after installation. No additional setup is required. It reads from the same checkin_entries and tcard_personnel database tables that are created during normal installation.

FEATURE	SETUP REQUIRED
ICS-211 check-in with Check Out button	None — available at /checkin.html immediately
QR/barcode scan check-in	None — available at /scan_checkin.html. Camera permission required on device.
PAR board with cross-reference	None — available at /accountability.html immediately
T-card personnel roster	None — Personnel tab appears on every T-card detail panel
Check-out recording	None — Check Out button on every check-in row in checkin.html and accountability.html

15.2 FEMA PA Cost Documentation

FEMA cost tracking is available immediately. The 2025 FEMA Schedule of Equipment Rates (44 categories) is pre-loaded during installation. Update rates annually at /fema_rates.html when FEMA publishes a new schedule.

FEATURE	NOTES
Force Account Labor with fringe calculation	Available immediately. ICS-214 import available from the Labor tab.
Equipment with FEMA rate schedule lookup	44 rates pre-loaded for 2025. Edit at /fema_rates.html annually.
Materials and Contracts tracking	Available immediately. Source documentation required for FEMA PW.
Project Worksheet text export	Click Export PW Text on any incident for a formatted PW summary.
Real-time cost dashboard	Available at /cost_dashboard.html. Auto-refreshes every 2 minutes.

15.3 Pre-Planned Event Templates

Six built-in templates are pre-loaded during installation: Emergency Shelter Operations, Search and Rescue, Severe Weather Response, Mass Gathering, HazMat Response, and Exercise/Training Scenario. Templates can be edited and new ones created at /event_templates.html.

STEP	ACTION
1	Navigate to http://192.168.50.1/event_templates.html
2	Click the **Activate** tab — browse the template gallery
3	Click a template — enter incident name and any custom details
4	Click **Activate Template** — a new incident is created pre-loaded with objectives, resource types, channels, and ICS organization structure
5	To edit a template: click **Manage Templates** → Edit → modify Objectives, Resources, Channels, or Organization tabs → Save Template
6	Export as JSON to share with other FieldCommand systems: click **Export JSON**

15.4 GPS-Tracked Resource Map

The GPS resource map (/resource_map.html) displays color-coded pins for all resources that have a GPS position set. Three placement methods are available.

METHOD	HOW TO USE
Device GPS (browser)	Click Set Position on any resource → click ■ Use Device GPS. Browser requests location permission. Accuracy shown in meters.
Click map to place	Click ■ Click Map to Place → crosshair cursor appears → click anywhere on the map to drop the resource pin.
Manual coordinates	Type decimal lat/lon directly into the fields. Negative longitude for western hemisphere.

15.5 IAP One-Click PDF Compilation

The IAP compiler (/iap_compile.html) generates a print-ready PDF containing all selected ICS forms for the active incident. Digital signatures are embedded. Generated server-side using ReportLab and pypdf — no browser PDF library needed.

STEP	ACTION
1	Ensure the active incident has completed ICS forms with signatures
2	Navigate to http://192.168.50.1/iap_compile.html
3	Check the forms to include (ICS-202, 203, 204, 205, 206, and 207 minimum)
4	Click ■ **Compile IAP PDF** — server generates the PDF (5–15 seconds)
5	PDF downloads automatically with title page, section dividers, and all forms
6	Print from any device using the Print Center (/printcenter.html)

15.6 NEXRAD Animated Radar

The radar page (/radar.html) requires an active internet connection for radar tile loading. When offline, it displays the last loaded frame with a timestamp. No installation configuration is required — it pulls tiles from NWS RIDGE II directly.

> ■ The radar page auto-detects WAN status and resumes loading new tiles automatically when internet connectivity is restored. No manual action is required after a WAN outage.

15.7 Dual-Source WAN Configuration

FieldCommand IMS supports two simultaneous WAN sources with automatic failover. Configure both sources at /wan_settings.html after installation.

FIELD	OPTIONS & DESCRIPTION
Role	Preferred (used first) · Fallback (used when preferred fails)
Type	Cellular · Satellite · Hotspot · Fixed broadband · Other
Detection method	internet_only — any internet path works (best for hotspots and USB dongles) | ping — checks a specific gateway IP | admin_reachable — checks modem admin interface URL
Label	Human-readable name shown on dashboard: "Cellular", "Satellite", etc.
Enabled	Toggle a source on or off without deleting configuration

STEP	ACTION
1	Navigate to http://192.168.50.1/wan_settings.html
2	Click **+ Add WAN Source** to add a source, or click Edit on an existing one
3	Set Role to **Preferred** for your primary source (e.g. cellular antenna)
4	Set Role to **Fallback** for your secondary source (e.g. satellite dish)
5	Choose the Detection Method appropriate for each source type
6	Click Save. The WAN monitor applies the new configuration within 60 seconds.
7	Use ■ **Swap Roles** to swap preferred and fallback without editing.

15.8 Digital Signature Capture

All 16 Prepared By and Approved By signature fields across the ICS form set support digital signature capture. Signatures are stored as PNG images in the database and are embedded in printed forms and compiled IAP PDFs. No setup required.

To sign: click any Prepared By or Approved By field → signature pad appears → sign with mouse, touchscreen, or stylus → click Accept. Signatures persist through page reloads and appear in the IAP PDF compilation.

15.9 Incident Archive and Beta Reset

FUNCTION	HOW TO USE	NOTES
Archive to USB	incident_mgmt.html → select incident → Archive to USB	USB drive must be labelled FIELDCOMMAND. JSON written to /media/fieldcommand/backup/incidents/
Restore from USB	incident_mgmt.html → Restore from USB → select JSON file	Restores incident to fully active status with all forms, T-cards, costs, and check-ins
Hard delete from Pi	incident_mgmt.html → select archived incident → Hard Delete	Permanent — confirm archive exists on USB before deleting
Beta/Scenario Reset	incident_mgmt.html → Beta/Scenario Reset → Confirm	Wipes all incidents, forms, T-cards, costs, check-ins. Preserves roster, channels, repeaters.

■ The USB drive must have volume label FIELDCOMMAND (all capitals). Format as FAT32 or exFAT. Label with: sudo mlabel -i /dev/sdX1 ::FIELDCOMMAND (replace sdX1 with actual USB device).

16. Maintenance & Updates

```
# Interactive maintenance menu:
sudo bash /opt/fieldcommand/scripts/update.sh

# Menu options:
# 1) Restart all services
# 2) Stop all services
# 3) Check service status
# 4) View live logs
# 5) Refresh FCC database
# 6) Fetch repeater data
# 7) Update web files from current directory
# 8) Backup data to /tmp
# 9) Show disk usage
```

USB Auto-Backup

Insert a USB drive formatted with the label **FIELDCOMMAND** to trigger an automatic backup of all runtime data. The backup copies /opt/fieldcommand/data/ to a timestamped folder on the USB drive.

```
# Format a USB drive with label FIELDCOMMAND (Linux):
sudo mkfs.vfat -n FIELDCOMMAND /dev/sda1
# Or on Windows: format as FAT32, set volume label to FIELDCOMMAND
# Backup destination on USB:
# /media/fieldcommand/backup/YYYYMMDD_HHMMSS
```

17. Troubleshooting

SYMPTOM	LIKELY CAUSE / FIX
http://192.168.50.1 not reachable	Pi not running, or ASUS router not configured with 192.168.50.x subnet. Check Pi power LED. Verify router LAN IP is 192.168.50.1.
Dashboard loads but cards give errors	Not connected to EMCOMM-NET — device is on a different network. Check Wi-Fi — must show EMCOMM-NET.
FCC lookup returns no results	FCC database not yet built. Run: sudo systemctl start fcc-refresh.service (needs internet)
APRS map shows no stations	Graywolf or YAAC not running, or no RF received yet. Check Health Monitor for service status. Confirm antenna connected.
Winlink form import fails	Wrong file type — must be the XML attachment, not the message body. In Winlink Express, right-click .xml attachment → Save As → then import.
Service dot is red on Health Monitor	Background service has stopped. SSH to Pi: sudo systemctl restart
Pat Winlink not accessible at port 8090	pat.service not running. Run: sudo systemctl start pat && sudo systemctl enable pat
JS8Call card does not open	TCP Server API not enabled in JS8Call, or wrong IP entered. Verify: File → Settings → Reporting → Enable TCP Server API, port 2442, hostname 0.0.0.0.
Printer not visible on EMCOMM-NET	avahi-daemon not running, or CUPS printer not shared. Check: sudo systemctl status avahi-daemon cups. Verify Share This Printer is checked in CUPS UI.
Issue: Services crash on startup	Low disk space or permission error. Check: df -h / and ls -la /opt/fieldcommand/. Verify fieldcommand user owns /opt/fieldcommand/.

■ For any issue not covered here — open the Health Monitor at http://192.168.50.1:5051/health or check the system log with: journalctl -u fcc-lookup -n 50

17. Quick Reference Card

ITEM	VALUE
FieldCommand dashboard	http://192.168.50.1
Wi-Fi network (SSID)	EMCOMM-NET
WAN status dashboard	http://192.168.50.1/wan-status.html
FieldCommand Pi static IP	192.168.50.1
the cellular antenna unit modem admin	http://10.1.1.1 or http://my.insty (connect to the cellular antenna unit Wi-Fi first)
the cellular antenna unit data plan	instyconnect.com — Multi-Network Unlimited — pause in portal between activations
satellite dish admin	http://192.168.10.1 (satellite dish app or browser)
44Net Gateway Pi IP	192.168.50.2
44Net gateway status	http://192.168.50.2:9000
AMPRNet tunnel check	ping 44.0.0.1 (from any EMCOMM-NET device)
ASUS router admin	http://192.168.50.254
CUPS printer admin	http://192.168.50.1:631
Pat Winlink (backup)	http://192.168.50.1:8090
Kiwix offline library	http://192.168.50.1:8081
Health monitor (raw)	http://192.168.50.1:5051/health
JS8Call TCP API (laptop)	http://[laptop-ip]:2442
Your callsign	your organization
your county grid	
Default coordinates	your deployment coordinates (decimal degrees)
Install log	/var/log/fieldcommand-install.log
Application data	/opt/fieldcommand/data/
HTML pages	/opt/fieldcommand/html/
Update script	sudo bash /opt/fieldcommand/scripts/update.sh